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Simulation-based dynamic optimization for ambulance dispatch and casualty distribution in mass casualty incident

Kuo-Hao Chang [a,b,c], Tzu-Li Chen [a,*], Lan-Hsin Yu [a], Tzu-Yin Chang [c]

[a] Department of Industrial Engineering and Engineering Management, National Tsing Hua University, Taiwan
[b] High Entropy Materials Center, National Tsing Hua University, Taiwan
[c] National Science and Technology Center for Disaster Reduction, Taiwan

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ABSTRACT

After an earthquake, the sudden influx of casualties from multiple zones overwhelms medical resources such as ambulances and hospital beds. This post-disaster mass casualty incident (MCI) is highly unpredictable, with conditions continuously evolving over time, making it urgent to develop real-time and flexible ambulance dispatch and casualty distribution strategies to maximize the expected patient survival rate. To address this, this study establishes a Markov Decision Process (MDP) model to make real-time decisions about ambulance dispatch, ambulance redeployment, and hospital selection based on the current state of the MCI system, optimizing resource allocation. Due to the high-dimensional and uncountable state space of this MDP model, and the inability to accurately estimate its transition probabilities, we propose a novel simulation-based approximate dynamic programming solution method, integrating a deep-neural-network-based approximate policy iteration (DNN-API) algorithm with a hybrid agent-based and discrete event simulation to efficiently generate near-optimal decisions. In collaboration with the National Science and Technology Center for Disaster Reduction in Taiwan, we conduct computational experiments using a potential earthquake scenario in Taipei City to assess the efficiency and effectiveness of our proposed simulation-based approximate dynamic programming algorithm compared to the least square time difference-based ADP method and current practice heuristics.

1. Introduction

In recent years, Taiwan has frequently experienced earthquakes, with an average of 20 to 30 earthquakes of magnitude 5 to 6 occurring annually. Earthquakes are unavoidable and each large-scale earthquake brings loss of life and property, causing buildings to collapse and leaving many people homeless. However, with proper planning, we can respond to disasters more effectively and reduce the damage. Disaster management can be divided into four stages: mitigation, preparation, response, and recovery, with the response phase being particularly critical. The goal of emergency management during this phase is to ensure the safety of affected individuals, provide timely rescue, and treat the injured to improve survival rates. Past research has mainly focused on post-disaster relief goods transportation, medical and livelihood resource distribution, and logistical support (Aslan and Çelik, 2019; Gao and Cao, 2020; Chang et al., 2022; Wang et al., 2023; Chang et al., 2024; Shi et al., 2024; Qezelbash-Chamak et al., 2024). However, there has been relatively little research on how to effectively dispatch ambulances and allocate hospital resources to improve survival rates when a large-scale earthquake occurs, especially when multiple regions are simultaneously overwhelmed with casualties, placing a heavy burden on ambulances and medical resources. Additionally, the destruction of infrastructure and traffic paralysis caused by the earthquake poses significant challenges to emergency responders in providing timely assistance. Therefore, to address these complex post-disaster scenarios, this research focuses on how to dynamically and flexibly dispatch ambulance resources to different disaster zones and allocate patients to suitable hospitals under highly uncertain and constantly changing post-disaster conditions, with the goal of significantly increasing patient survival rates.

In the response phase following a major earthquake disaster, a series of decision-making processes for mass casualty incidents will be initiated (as shown in Fig. 1), including strategic and operational decisions (Chang et al., 2023). Early tactical decisions in the post-disaster response phase include the CCP location decision, which determines the locations of casualty collection points (CCPs), and the EMS resource

* Corresponding author.
E-mail address: chentzuli@ie.nthu.edu.tw (T.-L. Chen).

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0305-0548/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

allocation decision, which allocates limited emergency medical services (EMS) resources based on the location of casualties and estimated casualty numbers. EMS resources, which are deployed to quickly mitigate the severity of potential threats, include ambulances, firefighters, and medical personnel. Subsequently, based on the CCP location and EMS resource allocation results, the following operational decisions will be made dynamically and iteratively over time to ensure that moderate and severe casualties are promptly transported to appropriate hospitals: (1) patient prioritization decision, which determines the order in which casualties are transported to hospitals; (2) hospital selection decision, which transfers severe casualties to appropriate hospitals via ambulances; (3) ambulance routing planning decision, which determines the shortest route to the hospital based on road damage; (4) ambulance redeployment decision, which re-deploys ambulances that have completed their service to CCPs with more casualties and greater need for EMS resources. This study aims to optimize these operational decisions based on the CCP locations and EMS resource allocation, also known as the Ambulance Dispatch and Casualty Distribution Problem, to minimize the number of deaths and maximize the number of survivors.

However, as shown in Fig. 1, the mass casualty system affected by operational decisions exhibits high uncertainty, including factors such as the number of arriving casualties, CCP treatment times, ambulance transport times, and even the self-medical treatment behavior of minor injury patients. Additionally, the status of the mass casualty environment will change dynamically over time. For example, during the ambulance transport process, a CCP may receive a large number of severely injured patients; hospital beds may be occupied by lightly injured patients, leading to resource shortages; or a moderate casualty at a CCP may deteriorate while waiting for an ambulance and become a severe injury requiring immediate treatment. Therefore, the static planning strategies generated by deterministic optimization models or two-stage stochastic programming models commonly used in the past literature cannot effectively solve the highly stochastic and dynamic ambulance dispatch and casualty distribution (SDADCD) problem. As a result, it is necessary to adopt a dynamic decision-making model to generate dynamic strategies, enabling decision-makers to make real-time ambulance dispatch and casualty distribution decisions based on the current situation of the mass casualty scene after the disaster.

In this paper, we first establish an MDP model to formulate the stochastic dynamic ambulance dispatch and casualty distribution problem. The model generates optimal strategies for ambulance dispatch, hospital selection, and ambulance redeployment, aiming to maximize the expected survival probability of casualties. Due to the high-dimensional and uncountable state space of this MDP model, as well as the inability to accurately estimate its transition probabilities, we further propose a novel simulation-based Approximate Dynamic Programming (ADP) solution method, which integrates deep-neural-network-based approximate policy iteration (DNN-API) algorithm with a hybrid agent-based discrete event simulation to efficiently generate near-optimal decisions. The DNN-API algorithm takes into account the spatial and temporal characteristics of the mass casualty system, constructs a set of basis functions to significantly reduce the high-dimensional and uncountable state space, uses them as input features for the deep neural network, and iteratively trains through a richer combination of nonlinear functions to effectively approximate the value function and improve the algorithm's performance. The high-resolution hybrid agent-based and discrete-event simulation model estimates the system state transition probabilities by capturing the stochastic dynamic behaviors of the overall mass casualty humanitarian logistics in greater detail. The model not only considers the stochastic and dynamic characteristics of mass casualty incident (MCI), including stochastic casualty arrival times, triage service times, random ambulance travel times, and dynamic changes in casualty conditions, but also incorporates agent-based modeling for the autonomous escape and self-medical behavior of minor injury patients. Through the iterative interaction between the DNN-API and the hybrid agent-based and discrete-event simulation model, an approximately optimal ambulance dispatch and casualty distribution strategy will be effectively generated. By collaborating with the National Science and Technology Center for Disaster Reduction (NCDR) in Taiwan, we conduct computational experiments using a potential earthquake scenario in Taipei City to assess the efficiency and effectiveness of our proposed simulation-based approximate dynamic programming algorithm, comparing it with the least square time difference (LSTD)-based ADP method and commonly used practical heuristics (e.g., Immediate-first and nearest-hospital-first heuristic,

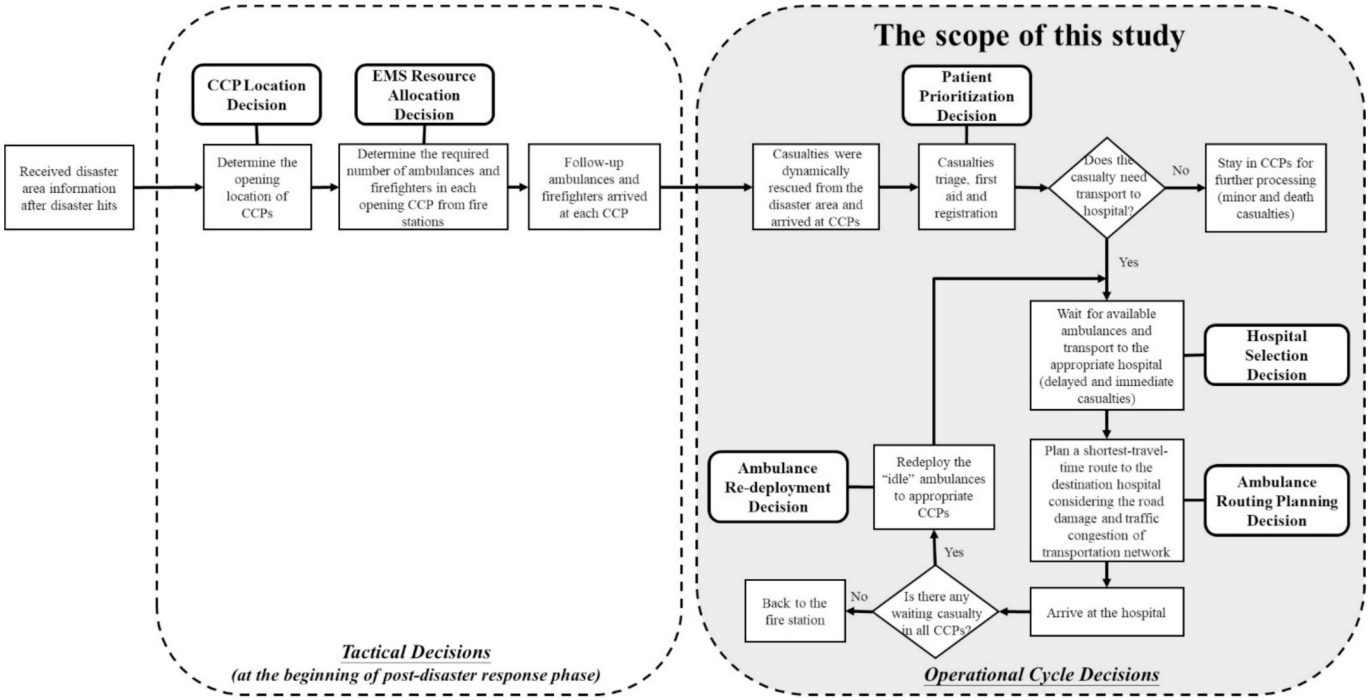


Fig. 1. Typical decision-making process for mass casualty incidents in the disaster response phase (Chang et al., 2023).

Immediate-first and cycle-hospital heuristic, etc.).

The remainder of this paper is organized as follows: Section 2 provides a literature review and highlights the research differences and gaps between previous studies and this paper. Section 3 formulates the Markov Decision Process model for the stochastic dynamic ambulance dispatch and casualty distribution problem. Section 4 introduces an innovative deep neural network-based approximate dynamic programming method to address the issues of the uncountable state space and the lack of analytical forms for the transition probabilities in the proposed MDP model. Section 5 presents a detailed description of the hybrid agent-based and discrete-event simulation model, inspired by an earthquake MCI scenario. Section 6 discusses the case study of a potential earthquake scenario in Taipei City, Taiwan, and presents the results of computational experiments. Finally, Section 7 provides the conclusion and suggestions for future research.

2. Literature review

2.1. Stochastic simulation and AI techniques

Stochastic simulation has long been used to analyze complex systems under uncertainty, particularly when random arrivals, stochastic service processes, congestion, resource interactions, and time-dependent dynamics make closed-form analysis difficult (Hong and Nelson, 2026). In dynamic stochastic systems, simulation is not only used to evaluate predefined policies, but also to support adaptive decision-making, because current decisions may affect future system states, resource availability, congestion, and long-term performance (Fu, 2002; Chang et al., 2013; Hong and Jiang, 2019). This feature creates a need to integrate stochastic simulation with optimization and artificial intelligence (AI) techniques, such as simulation optimization, approximate dynamic programming, reinforcement learning, and deep learning-based value-function approximation, to search for high-quality decisions or policies under uncertain and dynamically evolving environments (Fu, 2015; Powell, 2007; Sutton and Barto, 1999; Arulkumaran et al., 2017). Simulation optimization extends stochastic simulation from performance evaluation to decision search when objective functions and constraints can only be observed through noisy simulation outputs (Fu, 2002; Fu et al., 2015), while AI techniques provide learning and function-approximation capabilities for improving sequential decisions in large-scale and high-dimensional systems (Powell, 2007; Sutton and Barto, 1999; Arulkumaran et al., 2017). Recent studies further show that stochastic simulation can be combined with advanced computational and AI-inspired techniques to support large-scale decision-making, such as recurrent-neural-network-inspired simulation for inventory optimization and fast discrete-event simulation through Euler approximation (Wang and Hong, 2023; Hong et al., 2024). These studies indicate that stochastic simulation is no longer limited to ex post system evaluation, but has become a computational foundation for optimization, learning, and real-time decision support under uncertainty.

For dynamic sequential decision-making problems, simulation-based Markov decision process methods and approximate dynamic programming provide an important methodological bridge between stochastic simulation and artificial intelligence (AI) techniques. Classical dynamic programming becomes computationally infeasible when the state space, action space, or outcome space is large, continuous, or analytically intractable. To address this challenge, simulation-based MDP algorithms use simulation sampling to approximate transition dynamics, estimate value functions, and improve decision policies without full enumeration of all states and actions (Chang et al., 2005; Hu et al., 2007; Chang et al., 2013). Powell (2007) provides a comprehensive foundation for approximate dynamic programming and emphasizes that ADP is designed to address the "curses of dimensionality" arising from large state spaces, large action spaces, complex outcome spaces, and long decision horizons. In this sense, ADP is particularly useful for complex stochastic dynamic systems where exact dynamic programming is computationally infeasible (Powell, 2007; Chang et al., 2013).

Approximate dynamic programming is also closely related to reinforcement learning because both approaches learn decision policies from simulated or observed interactions with uncertain environments (Sutton and Barto, 1999; Powell, 2007; Chang et al., 2013). ADP typically emphasizes value-function approximation, policy evaluation, and policy improvement, while reinforcement learning provides a broader learning framework in which an agent improves its policy through repeated state-action-reward interactions (Sutton and Barto, 1999; Powell, 2007). Simulation-based MDP methods have also been extended to competitive or adversarial settings. For example, Chang et al. (2010) developed an adaptive adversarial multi-armed bandit approach for two-person zero-sum Markov games, showing that simulation-based learning can be used not only for single-agent MDPs but also for strategic dynamic decision-making problems. More recent Q-learning and relative Q-learning methods for continuous-state MDPs further demonstrate the applicability of learning-based ADP and reinforcement learning to practical operational systems with continuous and high-dimensional states (Hu et al., 2024; Yang et al., 2024).

Deep reinforcement learning further strengthens the integration between stochastic simulation and AI by using deep neural networks to approximate value functions, policies, or system representations in high-dimensional state spaces (Arulkumaran et al., 2017). From the perspective of stochastic simulation, DRL is important because simulation models can generate a large number of state-transition-reward trajectories for training, while deep neural networks can learn nonlinear mappings from system states to long-term values or actions (Arulkumaran et al., 2017). Recent studies have also extended this integration to risk-sensitive and multi-agent settings. Jiang et al. (2023) proposed quantile-based deep reinforcement learning using two-timescale policy gradient algorithms, showing that deep learning and reinforcement learning can be combined to optimize distributional or quantile-based performance criteria. Liu et al. (2025) applied multi-agent deep reinforcement learning to multi-echelon inventory management, demonstrating that AI-based learning can support decentralized and coordinated decision-making in complex stochastic operational systems with multiple interacting agents. These studies show that stochastic simulation can serve as a training environment for AI-based decision policies, while deep learning can provide the function-approximation capability needed to handle complex, high-dimensional, and nonlinear decision structures (Arulkumaran et al., 2017; Jiang et al., 2023; Liu et al., 2025).

Another important integration between stochastic simulation and AI is the offline simulation online application framework. Hong and Jiang (2019) proposed using extensive offline simulation experiments to generate training data for statistical or machine learning models that can be deployed online for fast decision support. Jiang et al. (2020) applied this framework to online risk monitoring, showing that simulation-generated data can be used to train predictive models for real-time risk classification. This framework highlights the complementary roles of stochastic simulation and AI: high-fidelity simulation can be used offline to capture complex system uncertainty and generate informative samples, while AI or statistical learning models can be used online to provide rapid prediction, classification, or decision support (Hong and Jiang, 2019; Jiang et al., 2020).

Recent studies further extend the integration of stochastic simulation and AI toward risk-aware and adaptive learning. Cao et al. (2025a) developed black-box CoVaR and gradient estimation methods for risk estimation in complex stochastic systems, while Cao et al. (2025b) proposed adaptive infinitesimal perturbation analysis derivative estimation with unknown parameters. These studies show that simulation-based methods can support risk estimation and sensitivity analysis even when system dynamics are black-box or input parameters must be learned from data (Cao et al., 2025a; Cao et al., 2025b). In addition, Sharpe ratio optimization in MDPs shows that simulation-based sequential decision-making can incorporate risk-adjusted performance

criteria beyond expected reward maximization (Ma et al., 2025). Bayesian stochastic gradient descent with streaming input data further supports adaptive stochastic optimization when input distributions are continuously updated by new observations (Liu et al., 2024), building on earlier foundations of streaming-data algorithms (O'Callaghan et al., 2002). These developments indicate that stochastic simulation and AI techniques are increasingly being integrated to address not only expected performance optimization, but also risk sensitivity, input uncertainty, and real-time data adaptation.

The present study contributes to this stream of research by developing an integrated application of stochastic simulation and AI techniques for dynamic ambulance dispatch and casualty distribution in mass casualty incidents. Specifically, the proposed method combines simulation-based approximate dynamic programming with a deep-neural-network-based approximate policy iteration algorithm to approximate the value function and learn high-quality dynamic dispatch policies. At the same time, a hybrid agent-based and discrete-event simulation model is used to capture stochastic system transitions, including casualty arrivals, triage processes, ambulance travel times, hospital-bed dynamics, and the autonomous self-medical behavior of minor-class patients. Therefore, this study represents a structured AI-enhanced stochastic simulation framework that integrates MDP modeling, ADP, deep value-function approximation, and hybrid simulation to support real-time ambulance dispatch, hospital selection, and ambulance redeployment decisions under highly uncertain and dynamically evolving post-disaster conditions.

2.2. Ambulance dispatch and casualty distribution in MCI

The dynamic dispatch and redeployment of ambulances for the treatment of mass casualty incidents have been discussed in previous literature. Maxwell et al. (2010) were the first to implement an ambulance redeployment policy to address the challenges of a busy emergency medical service system. They simulated ambulance redeployment as a dynamic program, dividing the system state into two parts: the state of the ambulances and the patients waiting for transportation to the hospital. Due to the generation of a high-dimensional and uncountable state space of this dynamic problem, an approximate dynamic programming was proposed to solve it. Methods for constructing parameter approximations of the update value functions include Temporal-Difference Learning (TD) and Linear Programming Approach. Finally, the algorithm was verified by applying it to the medical systems of two major cities.

Rettke et al. (2016) explored the issue of whether military medical personnel should consider dispatching air military medical evacuation during major combat operations. The authors developed a Markov decision process formulation and proposed an approximate policy iteration algorithm (API). They used the least squares time difference method (LSTD) to update the parameters of the approximate value function. Furthermore, they validated the model and assessed the algorithm's effectiveness by using it in a Syrian military operation.

Lam et al. (2017) pointed out that the demand for ambulances varies between weekdays and holidays. Therefore, they used a dynamic ambulance redeployment policy to enable more flexible resource allocation based on the current situation. They categorized the states into different discrete events and used three methods to make decisions: (1) determining the next destination for an ambulance after it completes its service, (2) updating the destination of the next trip of the ambulance at regular intervals, and (3) regularly updating the destination of the next trip for idle ambulances. After making these decisions, they used approximate dynamic programming to solve the problem. Finally, they validated the model and algorithm through the Singapore emergency medical system.

Mills et al. (2018) pointed out that after a large-scale incident, with limited ambulance resources, it is often the responsibility of the on-site commander to decide which hospital to send a patient to. To facilitate this decision, this study proposes a Markov decision process model and two heuristic algorithms to assist field commanders in decision. Considering the environment of multiple disaster areas and hospitals, the service time for triage follows an exponential distribution. The known factors include the average transportation time and the degree of hospital congestion. The decisions addressed by the model include (1) which ambulances are assigned to which casualty inspection stations, and (2) which hospitals treat patients transported by ambulances. The study considers the survival rate and service time of different grades of patients through simulation experiments. The results suggest that the heuristic algorithm significantly outperformed the traditional strategy of transporting to the nearest hospital.

Robbins et al. (2020) extended Rettke et al. (2016) to consider the sequential resource allocation decision of military medical evacuation of casualties during war. The author added the consideration of call control (admission control) into the model. Since the uncertainty of casualties and the service time increases the complexity of the model, they used a hierarchical aggregate value function within an approximate policy iteration algorithm to solve the model.

Shin and Lee (2020) pointed out that mass casualty events often lead to a shortage of emergency medical service resources such as ambulances and emergency departments, which must be managed effectively to save more lives. To address this, a stochastic dynamic model was developed that integrates patient transport priority and hospital selection problems. When comparing the strategies in the model with other heuristics, experimental results show that they outperform other alternatives. In addition, the research found that optimal hospital selection is as important as patient priority decisions. The study also proposed a heuristic strategy that considers patient priority and hospital selection. Experimental results show that this heuristic strategy performs and provides better policies than previous heuristic strategies.

Jenkins et al. (2021) extended Robbins et al. (2020) to study military medical evacuation during major combat operations. The main difference is that they incorporated air helicopter redeployment decisions into the model and used two algorithms to solve the problem. One is the approximate policy iteration algorithm based on the least square time difference method (LSTD-API), and the other is the approximate policy iteration algorithm based on the neural network method (NN-API). The results show that both algorithms achieve large improvements over usual strategies, with NN-API yielding slightly better results than LSTD-API.

Table 1 summarizes the research differences/gaps between the previous literature and the SDADCD problem investigated by this study. In addition, the major contributions of this paper are identified below.

(1) We consider the stochastic nature of the mass casualty system caused by a major earthquake disaster, including the uncertainty of ambulance travel time due to road disruptions, the types and number of casualties, CCP treatment times, and other factors. Additionally, we also account for the dynamic nature over time, including the deterioration of patient conditions, the dynamic evacuation of casualties to CCPs for treatment, and the occupancy of hospital beds by lightly injured patients seeking medical care autonomously. Based on this, we establish a Markov Decision Process model to solve the stochastic dynamic ambulance dispatch and casualty distribution decision problem and generate optimal strategies for ambulance dispatch, hospital selection, and ambulance redeployment, with the goal of maximizing the expected survival probability of casualties.
(2) We propose a novel simulation-based Approximate Dynamic Programming approach that combines deep neural network-based value function approximation (DNN-API) with a hybrid agent-based discrete-event simulation. This method efficiently tackles the challenges posed by the high-dimensional and uncountable state space in the proposed MDP model and the difficulty of accurately estimating its transition probabilities. The

Table 1
Summary of research differences/gaps between previous literature and this study.

Authors	Resource constraints						Stochastic factors	Solution methods	Disaster Scenarios
	Fire/ Ambulance stations	Hospital beds	Road network vulnerability	Multiple CCPs	Survival probability	Behavior of minor-class patients			
Maxwell et al. (2010)	✓						• Treatment time	• Linear dynamic programming	
Rettke et al. (2016)				✓	✓		• Travel time	• Approximate policy iteration based on least square temporal difference method algorithm	Aggression in northern Syria
Lam et al. (2017)	✓			✓			• Total number of patients	• Approximate policy iteration based on least square temporal difference method algorithm	
Mills et al. (2018)		✓		✓	✓		• The CCPs service time	• Markov decision process • Heuristic algorithm	Earthquake
Jenkins et al. (2021)				✓			• Travel time	• Approximate policy iteration based on least square temporal difference method algorithm • Approximate policy iteration based on neural network method algorithm	Military MEDEVAC planning in Azerbaijan
Robbins et al. (2020)				✓			• Total number of patients • Travel time	• Approximate policy iteration programming	Combat in Afghanistan
Shin and Lee (2020)	✓	✓		✓	✓		• The CCPs service time • Treatment time	• Markov decision process • Heuristic algorithm	Earthquake
This study	✓	✓	✓	✓	✓	✓	• Arrival time • The CCPs service time • Travel time • Number of hospital beds	• Markov decision process • Deep neural network-based value function approximation • Hybrid agent-based and discrete event simulation	Taiwan Earthquake

DNN-API algorithm incorporates the spatial and temporal dynamics of the mass casualty system by constructing a set of basis functions that significantly reduce the complexity of the state space. These basis functions are used as input features for the deep neural network, which is iteratively trained with a diverse combination of nonlinear functions to effectively approximate the value function and improve the algorithm's performance. Additionally, the high-resolution hybrid agent-based and discrete-event simulation model enhances the estimation of system state transitions by capturing the stochastic dynamic behaviors of mass casualty humanitarian logistics in greater detail. This model not only accounts for the stochastic and dynamic aspects of event-based mass casualty incidents but also integrates agent-based simulations to model the autonomous escape and self-medical behaviors of lightly injured patients. Through iterative interaction between the DNN-API and the hybrid agent-based simulation model, an approximately optimal strategy for ambulance dispatch and casualty distribution is derived.

(3) To demonstrate the effectiveness and practicality of our proposed model, we collaborate with the NCDR to conduct extensive numerical experiments using a potential earthquake scenario in Tainan City, Taiwan. The results show that our simulation-based ADP approach provides disaster management decision-makers with the most optimal and reliable ambulance dispatch and casualty distribution strategies. When compared to the least square temporal difference-based ADP method and four commonly used heuristics—Immediate-first and nearest-hospital-first heuristic (IF-NHF), Immediate-first and cycle-hospital heuristic (IF-CH), first-in-first-out and nearest-hospital-first heuristic (FIFO-NHF), and first-in-first-out and cycle-hospital heuristic (FIFO-CH)—our approach outperforms them in terms of efficiency and reliability. Through sensitivity analysis, we further investigate the impact of varying casualty arrival rates, available ambulance resources, available hospital bed resources, the probability of minor-class patients going to the hospital, and the degree of road damage caused by different earthquake magnitudes on the expected total survival probability of all casualties, as well as the robustness of the ambulance dispatch and casualty distribution strategies generated by the simulation-based ADP method.

3. Stochastic dynamic ambulance dispatch and casualty distribution problem

3.1. Problem description and definition

During a mass casualty incident, a significant number of patients require transportation to hospitals for treatment via ambulances. The number of ambulances and hospital beds is significantly lower compared to the number of patients, making them scarce resources. The planning of ambulance dispatch decisions directly impacts the survival rate of the patients. In addition, various uncertain factors contribute to changes in the state space, including the arrival time of the patients, the number of

available beds in the hospital, the travel time, and the service time of the CCPs. Due to the existence of various optimal dispatch decisions in the state space, we will construct a dynamic ambulance dispatch decision framework to determine ambulance dispatch, ambulance redeployment, and ambulance hospital selection decisions (as illustrated in Fig. 2). This study takes into account the above uncertain factors and uses the Markov decision process to construct a model. After the disaster occurs, information such as the number of ambulances in each fire station, the location of CCPs, the location of the disaster area, the location of the fire station, the location of the hospital, the probability of road damage, and the number of hospital beds in each hospital becomes known. It is assumed that ambulances start from the fire station, and all delayed-class and immediate-class patients are dispatched in ambulances. There is a fixed number of ambulances, and any patients who go to the hospital for medical treatment will be given treatment. The objective is to maximize the survival probability of the patients by dynamically altering the state information of the model and deriving the optimal decision. Furthermore, the stochastic simulation method is utilized to model a mass casualty incident system, aiding disaster responders in determining the appropriate ambulance dispatch strategy before a disaster strikes.

3.2. Markov decision process model

In this section, the stochastic dynamic ambulance dispatch and casualty distribution problem is formulated as a Markov Decision Process model. The indices and input parameters of the model are defined as follows and components of the proposed MDP model – system state, action, state transition, reward function and value function – are described in the following subsections.

Sets and Indices

I:	Set of fire stations, indexed by i
J:	Set of the CCPs, indexed by j
K:	Set of hospitals, indexed by k
L:	Set of ambulances, indexed by l
PI:	Patients classified as Immediate, indexed by PI
PD:	Patients classified as Delayed, indexed by PD
PM:	Patients classified as Minor, indexed by PM

Parameters

S:	The system state
$\boldsymbol{P}$:	The information of the CCPs
$\boldsymbol{Q}$:	The ambulance information
$\boldsymbol{H}$:	The hospital information
W:	A random element that captures the information arriving to the system
R:	The reward function
β:	The survival probability parameters values
γ:	A fixed discount factor
b_k:	The available number of beds for hospitals k
N_C:	The total available CCPs
N_A:	The total available ambulances
N_H:	The total available hospitals

3.2.1. State space

In the model, the system state will record the current time, CCP information, ambulances information, and hospitals information, which is defined as $S_t = (\tau_t, \boldsymbol{P}_t, \boldsymbol{Q}_t, \boldsymbol{H}_t)$ at decision epoch t, which τ_t represents the system current time. The decision epochs in the SDADCD problem refer to the points in time when mass casualty incident decision-makers are required to make ambulance dispatch and casualty transportation decisions. Let $T = \{1, 2, ...\}$ denote the set of decision epochs. A decision epoch is triggered when an ambulance completes the task of transporting a casualty to the hospital and its status changes from "busy" to "available". This decision epoch starts when a patient needs an ambulance (at this time $\tau_t = 0$) and continues until all the patients are sent to the hospital. The CCPs information is defined as $\boldsymbol{P}_t = (p_{t1}, p_{t2}, ..., p_{tN_C})$, $p_{tj} = \left(p_{tj}^{PI}, p_{tj}^{PD}, p_{tj}^{PM}\right), j = 1, ..., N_C$, where N_C is the total number of CCPs. For the information p_{tj} of each CCP at decision epoch t, p_{tj}^{PI}, p_{tj}^{PD} and p_{tj}^{PM} denote the number of immediate-class patients waiting for ambulances, the number of delayed-class patients, and the number of minor-class patients, respectively. The ambulance information is defined as $\boldsymbol{Q}_t = (q_{t1}, q_{t2}, ..., q_{tN_A})$, $q_{tl} = (q_{tl}^1, q_{tl}^2, q_{tl}^3, q_{tl}^4, q_{tl}^5, q_{tl}^6, q_{tl}^7)$, $l = 1, ..., N_A$, where N_A is the total number of ambulances. For the information q_{tl} of each ambulance at decision epoch t, we define 7-tuple to denote the attributes of an ambulance. q_{tl}^1 indicates whether the ambulance is available (if yes, $q_{tl}^1 = 0$, if not, $q_{tl}^1 = 1$). q_{tl}^2 denote the level/type of the patient the ambulance carries (if it is a delayed-class patient, $q_{tl}^2 = 1$, if it is an immediate-class patient, $q_{tl}^2 = 2$). q_{tl}^3 is the estimated time of arrival of the ambulance at the CCP. q_{tl}^4 is the estimated time when the ambulance arrives at the hospital. q_{tl}^5 records which CCP the ambulance is expected to arrive at. q_{tl}^6 records the destination hospital the ambulance is expected to arrive at. q_{tl}^7 records the arrival time of patient transported by ambulance l. The hospital information is defined as $\boldsymbol{H}_t = (h_{t1}, h_{t2}, ..., h_{tN_H})$, $h_{tk} = \left(h_{tk}^{PI}, h_{tk}^{PD}, h_{tk}^{PM}, b_{tk}\right), k = 1, ..., N_H$, where N_H is the total number of hospitals. $h_{tk}^{PI}, h_{tk}^{PD}, h_{tk}^{PM}$ and b_{tk} record the number of immediate-class patients already served, the number of delayed-class patients already served, the number of minor-class patients already served, and remaining beds in each hospital at decision epoch t respectively.

It is worth noting that the state space of the proposed MDP is not only high-dimensional but also uncountable. In general, the raw dimension of the state vector can be expressed as $1 + 3|J| + 7|A| + 4|H|$, where the first term represents the current system time, $3|J|$ corresponds to the three patient-count attributes for each CCP, $7|A|$ corresponds to the seven attributes describing each ambulance, and $4|H|$ corresponds to the four attributes associated with each hospital. For the small-scale case study in Section 6, where $|J| = 4$, $|A| = 5$ and $|H| = 4$, the raw state dimension is 64. For the large-scale case study, where $|J| = 16$, $|A| = 88$ and $|H| = 20$, the raw state dimension increases substantially to 697. These values already demonstrate the high dimensionality of the proposed MDP. Furthermore, the system state contains both discrete and continuous components. The discrete components include, for example, the numbers of immediate-, delayed-, and minor-class patients at each CCP, ambulance availability indicators, and the remaining bed capacities at hospitals. In contrast, the continuous components include the current decision time, the estimated arrival time of each ambulance at a CCP, the estimated arrival time at a hospital, and the arrival time of the casualty currently being transported. Since these time-related variables can take infinitely many real values and are continuously updated as the incident evolves, the overall state space is uncountable. Therefore, the proposed state representation forms a very large hybrid discrete–continuous state space, which makes exact dynamic programming computationally intractable.

3.2.2. Action space

The ambulance dispatching and casualty distribution decision is represented by the decision variable x_t at epoch t. At each decision epoch, if an ambulance l completes the transportation of the casualty to the hospital and its status changes from a busy to an available state ($q_{tl}^1 = 0$), the ambulance dispatching and casualty distribution decision is made decide which CCP it should go to, and which level of patient it should transport to which hospital for treatment. To capture dispatching and distribution decisions, we let $C(S_t) = \left\{j : j \in J, p_{tj}^{PI} > 0 \text{ or } p_{tj}^{PD} > 0\right\}$ denote the set of CCPs with patients waiting for rescue, $A(S_t) = \left\{l : l \in L, q_{tl}^1 \neq 1\right\}$ indicate the set of ambulance available for dispatching and $H(S_t) = \{k : k \in K, b_{tk} > 0\}$ denote the set of hospitals with remaining available beds. Let $x_{tjlk} = 1$ if the ambulance $l \in A(S_t)$ go to CCP $j \in C(S_t)$ to transport the "delayed-class patient" to the hospital $k \in H(S_t)$ when the system is in state S_t. Let $x_{tjlk} = 2$ if the ambulance $l \in A(S_t)$ go to CCP

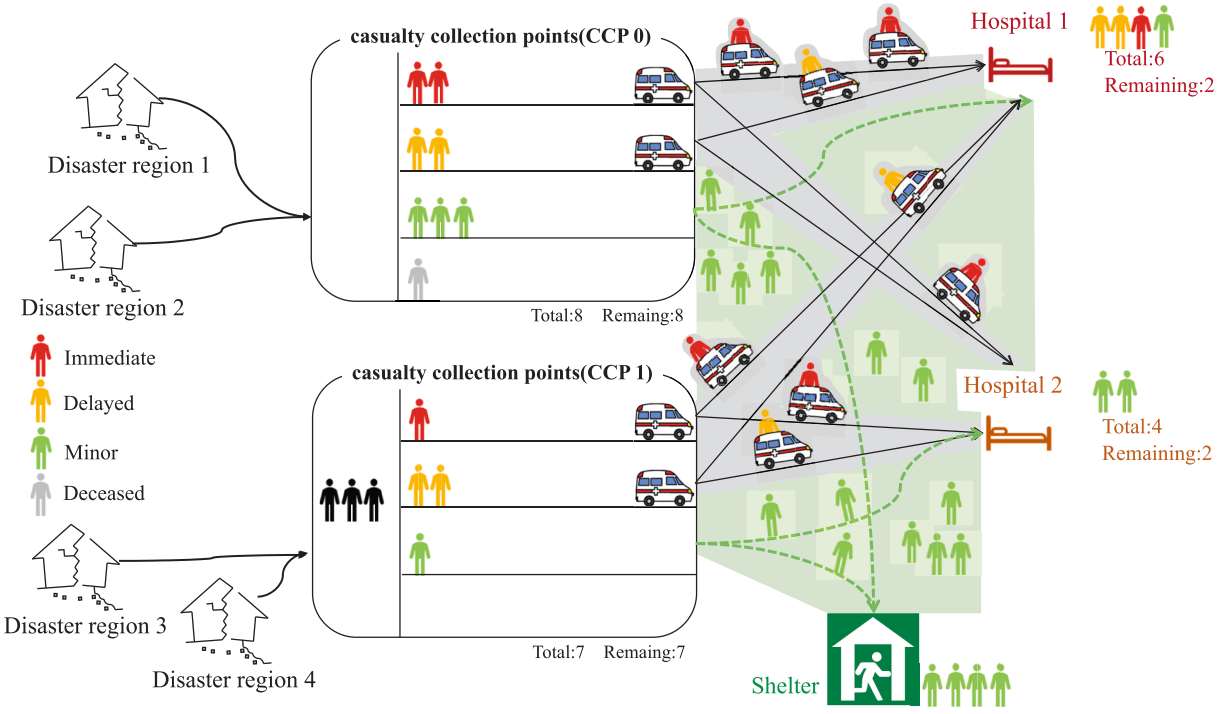


Fig. 2. Stochastic dynamic ambulance dispatch and casualty distribution problem.

$j \in C(S_t)$ to transport the "immediate-class patient" to the hospital $k \in H(S_t)$. If the ambulance $l \in A(S_t)$ is not dispatched, then $x_{tjlk} = 0$ for all $j \in C(S_t)$ and $k \in H(S_t)$. Therefore, the decision variable at decsion epoch t can be further denoted as $x_t = \{x_{tjlk} : j \in C(S_t), l \in A(S_t), k \in H(S_t)\}$. The set of feasible decisions can then be written in Equation (1).

$$\chi(S_t) = \left\{ x_t \in \{0,\ 1,\ 2\}^{|C(S_t)|\times|A(S_t)|\times|H(S_t)|} \right\} \tag{1}$$

As with the state space, the action space of the proposed MDP is also high-dimensional. In general, the upper-bound dimension of the action vector can be expressed as $2|A||J||H|$, where the factor 2 corresponds to the two patient severity classes considered in the model, namely immediate-class and delayed-class patients, $|A|$ denotes the number of available ambulances, $|J|$ denotes the number of CCPs, and $|H|$ denotes the number of hospitals. For the small-scale case study in Section 6, where $|A| = 5$, $|J| = 4$ and $|H| = 4$, the upper-bound action dimension is 160. For the large-scale case study, where $|A| = 88$, $|J| = 16$ and $|H| = 20$, the upper-bound action dimension increases substantially to 56,320. These values clearly demonstrate the high dimensionality of the action representation. Moreover, because each feasible action must jointly determine the ambulance assignment, the target CCP, the patient type, and the destination hospital under the current system conditions, the feasible action space is combinatorial in nature, which further increases the computational complexity of solving the proposed MDP exactly.

3.2.3. State transitions

State transitions describe how a mass casualty incident system updates from one state to another as new information arrives and decisions are made. The state transition function $S_{t+1} = S^M(S_t, x_t, W_{t+1})$ represents a dynamics of mass casualty incident system. The state S_{t+1} of the system at the beginning of epoch $t+1$ is determined based on the state S_t of the system at epoch t, the decision x_t that is made at epoch t and the arrival of new information W_{t+1} at epoch $t+1$. However, due to the high complexity and dynamic nature of the MCI system, it is not feasible to provide a closed-form expression of the state transition function. Therefore, this study develops a hybrid agent-based and discrete event simulation model to approximate and estimate the state transition function in Section 5.

3.2.4. Reward function

In our model, the reward is defined as the survival probability of a patient and is granted when the patient arrives at the hospital and begins receiving treatment. During a mass casualty incident, the treatment of a large number of casualties may be hindered by shortages of ambulances and medical resources, leading to treatment delays and, consequently, lower survival probabilities. Therefore, we adopt patient survival probability as the reward function, since earlier transportation and treatment generally lead to a higher probability of survival. As illustrated in Fig. 3, the survival probability decreases as the time from system entry to hospital treatment increases. Fig. 3 shows two survival probability curves, namely the solid curve for immediate-class patients and the dashed curve for delayed-class patients. The immediate-class curve lies below the delayed-class curve and declines more rapidly over time, reflecting the fact that immediate-class patients are in more critical condition and are more sensitive to treatment delay. In contrast, delayed-class patients are relatively less severe, so their survival

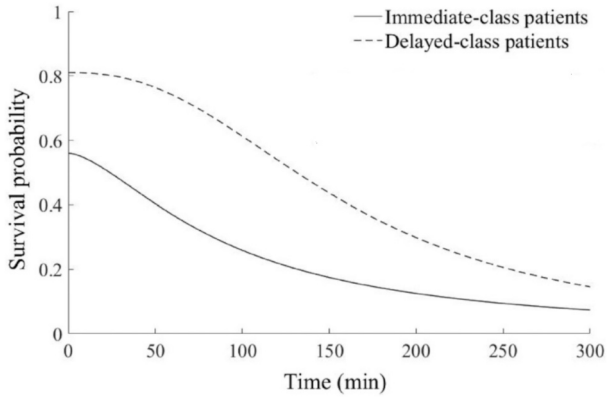


Fig. 3. Survival probability function.

probability starts at a higher level and decreases more gradually as treatment is delayed. This distinction further highlights the importance of prioritizing immediate-class patients in ambulance dispatch and hospital transportation decisions. Equation (2) represents the survival probability functions for different injury severity classes. This study considers only immediate-class patients (PI) and delayed-class patients (PD). The parameters $\beta_{0,k}$, $\beta_{1,k}$ and $\beta_{2,k}$ denote the coefficients of the survival probability functions for the patient type k, and these coefficients are set according to the recommendations of Mills et al. (2013). Once a patient is transported to the hospital and begins receiving treatment, the reward function in Equation (3) calculates the corresponding survival probability based on the time difference $(q_{tl}^4 - q_{tl}^7)$, where q_{tl}^4 is the time when the patient is transported by an ambulance to the hospital and starts receiving treatment and q_{tl}^7 represents the time point when the patient enters the system.

$$SF^k(t) = \frac{\beta_{0,k}}{\left(\frac{t}{\beta_{1,k}}\right)^{\beta_{2,k}} + 1} \quad k = PI, PD \tag{2}$$

$$R_t(S_t, x_t) = \begin{cases} SF^{PD}(q_{tl}^4 - q_{tl}^7), \text{ if } \tau_t = q_{tl}^4 \text{ and } q_{tl}^2 = 1 \text{ (delayed-class patients)} \\ SF^{PI}(q_{tl}^4 - q_{tl}^7), \text{ if } \tau_t = q_{tl}^4 \text{ and } q_{tl}^2 = 2 \text{ (immediate-class patients)} \\ 0, \text{ otherwise.} \end{cases} \tag{3}$$

3.2.5. Objective function and optimality equation

We define $X^\pi(S_t)$ as the decision function that, under a given policy π, returns an action x_t under the state S_t. Our MDP model aims to identify the optimal policy π^* from the set of all possible policies $(X^\pi(S_t))_{\pi\in\Pi}$ that maximizes the expected total discounted reward accumulated by the mass casualty incident system. Thus, we write the objective of the MDP in Eq. (4).

$$\max_{\pi\in\Pi} \mathbf{E}^\pi \left[\sum_{t=1}^{T} \gamma^t \times R_t(S_t, X^\pi(S_t)) \right] \tag{4}$$

where $\gamma \in (0, 1]$ is a fixed discount factor and T denotes the time at which the last patient is transported to the hospital. We can also use the following Bellman's optimality Equation (5) to find the optimal policy π^*.

$$V_t(S_t) = \max_{x_t \in \chi(S_t)} (R_t(S_t, x_t) + \gamma \times \mathbf{E}[V_{t+1}(S_{t+1}) | S_t, x_t]) \tag{5}$$

Unfortunately, the proposed MDP model has a high-dimensional and uncountable state space, and the state transition function lacks a closed-form expression, making it highly intractable to compute the optimal policy using Equation (5). Therefore, in the following sections, this study instead proposes a novel simulation-based approximate dynamic programming solution method to obtain high-quality ambulance dispatch and casualty distribution decision policy.

4. Solution methodology

To address the complex and large-scale MDP model proposed in this study, we further develop a novel simulation-based approximate dynamic programming solution method to construct high-quality and near-optimal ambulance dispatch and casualty distribution policies. As shown in Fig. 4, this simulation-based ADP approach consists of two components: (1) a deep-neural-network-based approximate policy iteration (DNN-API) algorithm, and (2) a hybrid agent-based and discrete event simulation. In the DNN-API, the optimality equation (5) of the MDP model is first reformulated into a post-decision-state-based formulation to eliminate the challenge of computing the embedded expectation in the original optimality equation. Subsequently, the DNN-API preserves the two fundamental phases of the traditional policy iteration (PI) algorithm—policy evaluation and policy improvement. In contrast to the conventional PI approach, DNN-API incorporates a stochastic simulation model during the policy evaluation phase to overcome the curse of dimensionality caused by the need to enumerate all possible states, while also capturing the dynamic nature of state transitions. Moreover, a deep neural network is utilized to approximate the post-decision state value function over the high-dimensional and uncountable state space. In the policy improvement phase, the simulation data collected during policy evaluation are used to retrain the deep neural network, thereby yielding improved decision policies. To reduce the dimensionality of the high-dimensional and uncountable state space

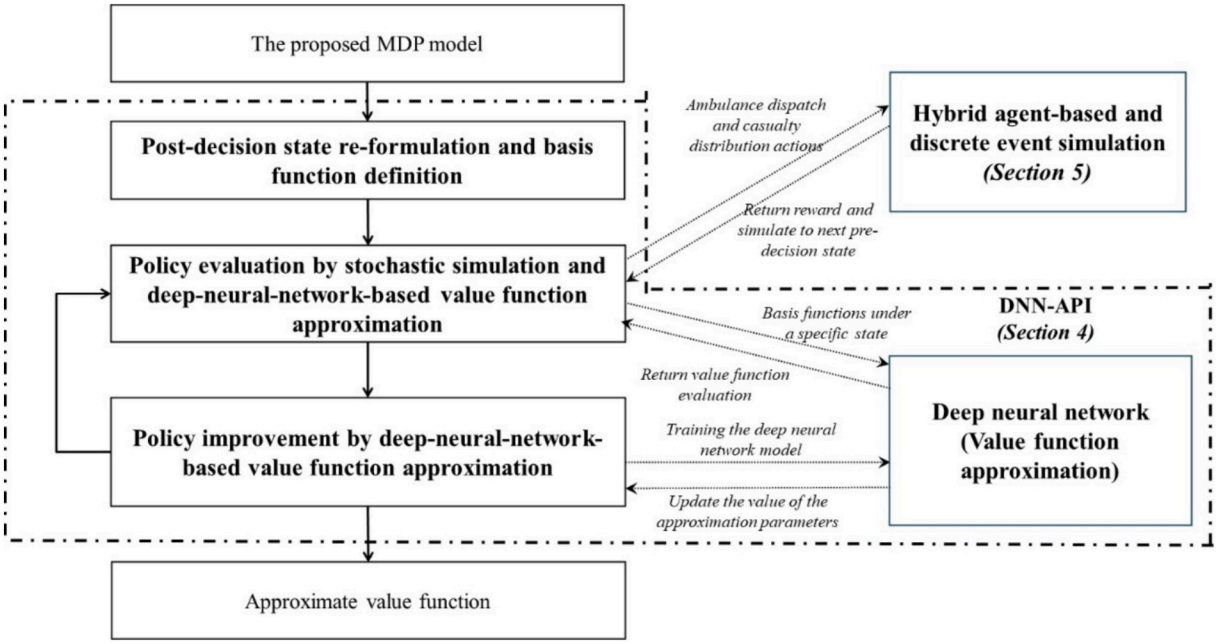


Fig. 4. A novel simulation-based approximate dynamic programming with approximate policy iteration, deep neural network, and hybrid agent-based and discrete event simulation.

in the proposed MDP model, we further identify and construct a fixed set of basis functions specific to the ambulance dispatch and casualty distribution problem. These basis functions serve as key input features to the deep neural network for approximating the post-decision state value function. The detailed implementation of the DNN-API algorithm will be presented in the following subsections. The hybrid agent-based and discrete event simulation serves a dual purpose: it captures the stochastic and dynamic behaviors of the MCI system to generate state transitions for policy evaluation, and it incorporates agent-based modeling to simulate the self-initiated hospital visits of minor-class patients, which in turn dynamically impact the availability of hospital beds. Section 5 provides a detailed explanation of the hybrid agent-based and discrete event simulation model.

The proposed DNN-API is closely related to deep reinforcement learning (DRL) in that it combines an MDP formulation with a deep neural network to approximate the value function in a high-dimensional and uncountable state space. In this sense, it shares the same fundamental motivation as value-based DRL methods such as DQN, which use neural networks to represent value functions in complex sequential decision problems (Mnih et al., 2015; Sutton and Barto, 2018). However, our approach is not a standard end-to-end model-free DRL algorithm. Instead, it is more appropriately viewed as a simulation-based approximate dynamic programming / approximate policy iteration method with deep value-function approximation (Bertsekas and Tsitsiklis, 1996; Bertsekas, 2001). While both ADP and DRL aim to solve sequential decision problems under uncertainty, ADP is rooted in approximate solution methods for dynamic programming and typically emphasizes value-function approximation and policy iteration, whereas DRL is more commonly used as a broader learning paradigm that combines reinforcement learning with deep neural networks through repeated interaction with the environment (Bertsekas and Tsitsiklis, 1996; Sutton and Barto, 2018). In particular, the proposed method relies on a post-decision-state reformulation, uses problem-specific basis functions as input features, and performs policy evaluation through a hybrid agent-based and discrete-event simulation model. Therefore, the proposed framework may be regarded as a structured deep approximate dynamic programming method that is conceptually related to DRL, while being specifically tailored to the stochastic ambulance dispatch and casualty distribution problem.

4.1. Post-decision state formulation

The high dimensionality and uncountable nature of the state space in the proposed MDP model make solving Equation (5) using exact dynamic programming techniques, such as value iteration or policy iteration, computationally infeasible. To address this challenge, this study employs an approximate dynamic programming (ADP) framework that focuses on approximating the value function with respect to the post-decision state variable (Powell 2007). The post-decision state formulation provides two key computational advantages: first, it simplifies the value function structure, effectively avoiding the challenges and complexities associated with computing the embedded expectation in the original optimality equation (5); second, it facilitates the use of stochastic simulation and sampling methods to substantially reduce the dimensionality of the state space (Powell 2007, Ruszczyński, 2010). A post-decision state S_t^x captures the state of the system immediately after making a decision x_t based on the pre-decision state S_t but before any new information W_{t+1} in a mass casualty incident system has arrived at epoch $t+1$. Utilizing this information, the value of being in the post-decision state S_t^x is denoted by $V^x(S_t^x)$. Then, the relationship between $V_{t+1}(S_{t+1})$ and $V^x(S_t^x)$ is given by Equation (6).

$$V^x(S_t^x) = \mathbf{E}\left[V_{t+1}(S_{t+1})|S_t^x\right] \tag{6}$$

By substituting Equation (6) into Equation (5), the Bellman's optimality Equation (5) is modified and reformulated as follows.

$$V_t(S_t) = \max_{x_t \in \chi(S_t)} \left(R_t(S_t, x_t) + \gamma \times V^x(S_t^x)\right) \tag{7}$$

Similarly, the value of being in post-decision state S_{t-1}^x is also expressed as follows.

$$V^x(S_{t-1}^x) = \mathbf{E}\left[V_t(S_t)|S_{t-1}^x\right] \tag{8}$$

By substituting Eq. (7) into Eq. (8), the optimality equation or value function of the post-decision state variable S_t^x can be reformulated and obtained as follows.

$$V^x(S_{t-1}^x) = \mathbf{E}\left[\max_{x_t \in \chi(S_t)} \left(R_t(S_t, x_t) + \gamma \times V^x(S_t^x)\right)|S_{t-1}^x\right] \tag{9}$$

Although the use of the post-decision state formulation brings computational advantages, solving the optimality Eq. (9) remains computationally intractable due to the size and dimensionality of our motivating problem instance. One of the most effective and widely adopted approaches to address complex dynamic programming problems is to replace the true value function with a statistical approximation. In this study, a set of basis functions is designed and identified to capture the key features of the post-decision state variables associated with the ambulance dispatch and casualty distribution problem in the MCI system. These basis functions are then integrated with a deep neural network to build an approximation model for the post-decision state value function (9).

4.2. Basis functions

To approximate the post-decision state value function in the ambulance dispatch and casualty distribution problem, selecting appropriate basis functions and features is a critical and challenging task. It is essential for deriving high-quality ambulance dispatching policies and effectively addressing the curse of dimensionality. This study leverages the work of Maxwell et al. (2010) and Jenkins et al. (2021), and incorporates the specific characteristics of the SDADCD problem. By adopting extensive trial-and-error experimentation and a stepwise-type feature selection procedure (e.g., forward selection or backward elimination etc.), we identify and design ten representative and effective basis functions. These functions are used as input features to the deep neural network for approximating the post-decision state value function. We let $\phi_f(S_t^x)$ be the basis functions, where $f \in F$ is the feature, and F is the set of features. These ten basis functions are defined as follows.

$$\phi_{1j}(S_t^x) = p_{tj}^{PI}, \forall j \in J \tag{10}$$

Eq. (10) describes how many immediate-class patients are to be transported from each CCP j in the system.

$$\phi_{2j}(S_t^x) = p_{tj}^{PD}, \forall j \in J \tag{11}$$

Eq. (11) describes how many delayed-class patients are to be transported from each CCP j in the system.

$$\phi_{3j}(S_t^x) = p_{tj}^{PM}, \forall j \in J \tag{12}$$

Eq. (12) describes how many minor-class patients are to be transported from each CCP j in the system.

$$\phi_{4l}(S_t^x) = q_{tl}^1, \forall l \in L \tag{13}$$

Eq. (13) indicates whether each ambulance l is an available ambulance (if yes, $q_{tl}^1 = 0$, if not, $q_{tl}^1 = 1$).

$$\phi_{5l}(S_t^x) = q_{tl}^2, \forall l \in L \tag{14}$$

Eq. (14) indicates what level of patients each ambulance l should transport (if it is a delayed-class patient, $q_{tl}^2 = 1$, if it is an immediate-class patient, $q_{tl}^2 = 2$).

$$\phi_{6l}(S_t^x) = q_{tl}^3, \forall l \in L \tag{15}$$

Eq. (15) represents the estimated arrival time of each ambulance l at the CCP.

$$\phi_{7l}(S_t^x) = q_{tl}^4, \forall l \in L \tag{16}$$

Eq. (16) represents the estimated arrival time of each ambulance l at the hospital.

$$\phi_{8k}(S_t^x) = h_{tk}^{PI}, \forall k \in K \tag{17}$$

Equation (17) describes the number of immediate-class patients in each hospital k in the system.

$$\phi_{9k}(S_t^x) = h_{tk}^{PD}, \forall k \in K \tag{18}$$

Eq. (18) describes the number of delayed-class patients in each hospital k in the system.

$$\phi_{10k}(S_t^x) = b_{tk}, \forall k \in K \tag{19}$$

Eq. (19) describes how many beds are left in each hospital k in the system.

4.3. Deep-neural-network-based value function approximation

In this section, we develop a nonlinear post-decision state value function approximation method based on deep feed-forward neural networks by incorporating the basis functions introduced in the preceding section. The deep feed-forward neural network used to approximate the post-decision state value function $V^x(S_t^x)$ consists of $2+G$ layers, including an input layer, G hidden layers, and an output layer, as illustrated in Fig. 5. The input layer contains F nodes, where each node f receives the evaluated value of the corresponding basis function $\phi_f(S_t^x)$ under the post-decision state S_t^x. In the G hidden layers, each hidden layer g contains a set of activation nodes $D = \{1, 2, ..., |D|\}$ (i.e., nonlinear perceptron nodes). We define $\Theta^1 = \left[\theta_{f,d}^{(0)}\right]_{f\in F, d\in D}$ is an $|F| \times |D|$ matrix of weights controlling the function mapping from the input layer to the first hidden layer ($g = 1$). Because first hidden layer requires $|D|$ input values, the inputs of first hidden layer are defined in Equation (20).

$$Y_d^{(1)}(S_t^x) = \sum_{f=1}^{|F|} \theta_{f,d}^{(0)} \times \phi_f(S_t^x), \forall d \in D \tag{20}$$

Next, we adopt the following nonlinear logistic sigmoid function as the activation function (21).

$$\sigma(y) = \frac{1}{1 - e^{-y}} \tag{21}$$

By applying activation function (21) to $Y_d^{(1)}(S_t^x)$, the output of the first hidden layer is given by the following expression in Eq. (22).

$$Z_d^{(1)}(S_t^x) = \sigma\left(Y_d^{(1)}(S_t^x)\right), \forall d \in D \tag{22}$$

Similarly, we define $\Theta^2 = \left[\theta_{d,d}^{(g)}\right]_{g=1,2,...,G-1, d\in D, d\in D}$, is an $(G-1) \times |D| \times |D|$ matrix of weights controlling the function mapping from the gth hidden layer to $(g+1)$th hidden layer. Based on this weight matrix, the inputs and outputs of each hidden layer g from $g = 2$ to $g = G$ are given by the following Eqs. (23) and (24), respectively.

$$Y_d^{(g)}(S_t^x) = \sum_{d=1}^{D} \theta_{d,d}^{(g-1)} \times Z_d^{(g-1)}(S_t^x), \forall d \in D, g = 2, 3, ..., G \tag{23}$$

$$Z_d^{(g)}(S_t^x) = \sigma\left(Y_d^{(g)}(S_t^x)\right), \forall d \in D, g = 2, 3, ..., G \tag{24}$$

Finally, we define $\Theta^3 = \left[\theta_d^{(G)}\right]_{d\in D}$ is an $|D| \times 1$ matrix of weights controlling the function mapping from the Gth hidden layer to output layer. Based on this weight matrix, both the input and output of the output layer are single scalar values, defined respectively by Eqs. (25) and (26). Eq. (26) also represents the computation results in the post-decision state value function approximation under the post-decision state S_t^x.

$$Y^{(Final)}(S_t^x) = \sum_{d=1}^{D} \theta_d^{(G)} \times Z_d^{(G)}(S_t^x) \tag{25}$$

$$\overline{V}^x(S_t^x|\boldsymbol{\theta}) = \sigma\left(Y^{(Final)}(S_t^x)\right) \tag{26}$$

wherein $\boldsymbol{\theta} = (\Theta^1, \Theta^2, \Theta^3)$ is a parameter tuple that succinctly represents the weights of the deep neural network (DNN), and it is trained and updated using the data collected through simulation sampling in the DNN-API framework. In addition, for a given parameter tuple $\boldsymbol{\theta}$, which represents a fixed and specific policy π, Eq. (27) can be used to make the decisions based on this policy during the policy evaluation phase of the

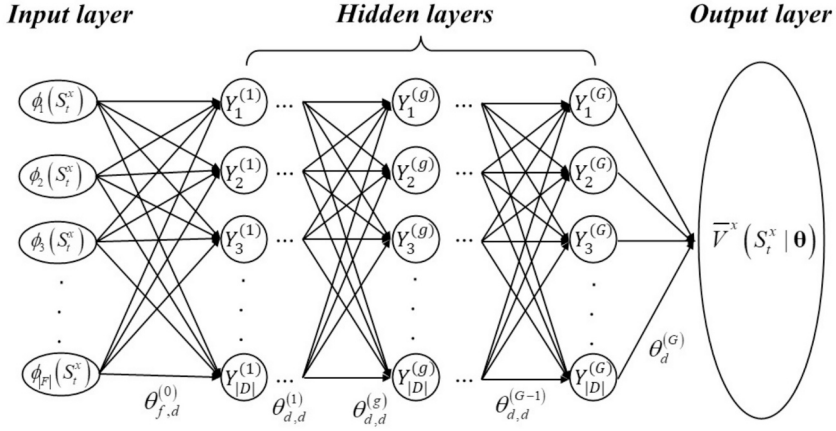


Fig. 5. Deep feed-forward neural network used to approximate the post-decision state value function.

proposed DNN-API algorithm.

$$X^{\pi}(S_t|\boldsymbol{\theta}) = \underset{x_t \in \chi(S_t)}{\operatorname{argmax}}\left(R_t(S_t, x_t) + \gamma \times \overline{V}^{x}\left(S_t^{x}|\boldsymbol{\theta}\right)\right) \tag{27}$$

4.4. DNN-API algorithm

This subsection develops the DNN-API algorithm, which integrates basis functions, a deep-neural-network-based value function approximation, and a hybrid agent-based and discrete event simulation model to derive approximately optimal strategies for ambulance dispatch and casualty distribution. The pseudocode of the DNN-API algorithm is presented in Table 2. The DNN-API algorithm starts by specifying the weight decay (η) and step-size (α_1) parameters and initializing the DNN weight parameters $\boldsymbol{\theta} = \left(\Theta^1, \Theta^2, \Theta^3\right)$, which represent a fixed initial policy. Following the recommendation of Hastie et al. (2009, Sec. 11.4), the initial weights are set to small random values close to zero. We then begin the policy evaluation phase, during which the following steps are performed for each iteration $m = 1, \ldots, M$. Within each iteration, total discounted reward of the MCI system is collected by simulating the trajectory of a stationary policy x_t, derived from the current DNN-based post-decision state value function approximation $\overline{V}^{x}\left(S_t^{x}|\boldsymbol{\theta}\right)$, over a set period T of time. At the beginning of each iteration m, we randomly select a post-decision state $S^x(n,m)$ and generate the corresponding next pre-decision state $S_{t=1}$ associated with it through the hybrid agent-based and discrete event simulation model. Utilizing the DNN-based post-decision state value function approximation and Eq. (30), we compute the optimal action x_t under the current pre-decision state S_t and the immediate reward $R_t(S_t, x_t)$ and the corresponding next post-decision state S_t^x obtained from executing this optimal action. Next, we simulate the system evolving from the post-decision state S_t^x to next pre-decision state S_{t+1} by running the hybrid agent-based and discrete event simulation model. This loop runs for a total of T periods, that is, until all casualties have been transported to the hospital for treatment. We also calculate the total discounted reward, $\widehat{v}(n,m) = \sum_{t=1}^{T} \gamma^{t-1} \times R_t(S_t, x_t)$, from the initial post-decision state $S^x(n,m)$ until all patients have been transported to the hospital at the decision epoch T. For each policy evaluation phase, we collect a set M of sample realizations of the value $\widehat{v}(n,m)$ achieved by following the current policy across different post-decision states $S^x(n,m)$.

Upon completion of the policy evaluation phase, the algorithm proceeds to the policy improvement phase wherein, for each iteration $n = 1, \ldots, N$. We use the M value function sample realizations collected during the policy evaluation phase to train the deep neural network model and update its weights $\boldsymbol{\theta}$, enabling DNN model to accurately fit the collected approximate value function data and obtain an updated policy. We adopt a regularized mean-squared error measure as the loss function, as shown in Eq. (28). The penalty term in the loss function (28) serves to prevent overfitting and reduce generalization error. In this study, we employ the L^2 penalty because it corresponds to the standard weight-decay regularization widely used in neural networks, which improves generalization by smoothly shrinking the weights in $\boldsymbol{\theta}$ toward zero (Bishop, 2006; Goodfellow et al., 2016). Since the proposed DNN model uses a compact set of representative basis functions as input features to approximate the post-decision state value function, our objective is to obtain a stable and smooth value-function approximation rather than to induce sparsity in the parameter estimates. Compared with alternative penalty functions such as the L^1 penalty, which is more commonly associated with sparse estimation and variable selection (Zou and Hastie, 2005), the L^2 penalty is differentiable and yields a smoother optimization landscape, making it more suitable for the gradient-based and second-order training procedures adopted in this study. The regularization parameter $\eta \geqslant 0$ (also known as weight decay) is a tunable hyperparameter; higher values of η encourage the weights in $\boldsymbol{\theta}$ to shrink toward zero. To enable the DNN model to approximate the value function and obtain updated weights $\boldsymbol{\theta}$, the loss function must be minimized (Equation (29)). This can be achieved using various computational techniques, such as the backpropagation algorithm, Newton's method, or the trust-region method (Hastie et al., 2009; Jenkins et al., 2021), to generate a new sample estimate $\boldsymbol{\theta}^{*}$ of $\boldsymbol{\theta}$.

Table 2
Pseudo code of DNN-API algorithm.

1. Initialize the DNN weights $\boldsymbol{\theta}$ with small random values close to zero, weight decay η, step-size α_1, initial exploration probability ε_0, minimum exploration probability $\varepsilon^{\min}$ and decay factor ψ
2. **For** $n = 1, \ldots, N$(Policy improvement loop)
3. **Set** $\varepsilon_n = \max\left\{\varepsilon^{\min}, \varepsilon_n \times \psi^n\right\}$
4. **For** $m = 1, \ldots, M$(Policy evaluation loop through the hybrid agent-based and discrete event simulation model)
5. Initialize the hybrid agent-based and discrete event simulation configuration
6. Generate an initial post-decision state $S^x(n,m)$ randomly
7. Generate next pre-decision state associated with $S^x(n,m)$ and call it $S_{t=1}$ by running the hybrid agent-based and discrete event simulation model
8. **For** $t = 1, \ldots, T$(T denotes the time the last patient reaches the hospital)
9. Generate a random number $u \sim \mathit{Uniform}(0,1)$
10. **If** $u > \varepsilon_n$ then
11. Generate a greedy action x_t by using $x_t = \underset{x_t \in \chi(S_t)}{\operatorname{argmax}}\left(R_t(S_t, x_t) + \gamma \times \overline{V}^{x}\left(S_t^{x}|\boldsymbol{\theta}\right)\right)$
12. **Else**
13. Generate an exploratory action x_t by randomly selecting a feasible action from $\chi(S_t)$
14. **End if**
15. Compute and record the immediate reward $R_t(S_t, x_t)$ associated with action x_t taken in current pre-decision state S_t
16. Obtain the next post-decision state S_t^x using the action x_t and current pre-decision state S_t
17. Run the hybrid agent-based and discrete event simulation model to obtain next pre-decision state S_{t+1} at the decision epoch $t+1$
18. **End for**
19. Calculate total discounted reward $\widehat{v}(n,m)$ incurred starting from the given state $S^x(n,m)$ by using $\widehat{v}(n,m) = \sum_{t=1}^{T} \gamma^{t-1} \times R_t(S_t, x_t)$
20. **End for**
21. Compute the tunable parameters $\boldsymbol{\theta}^{*}$ at the next iteration by minimizing the following loss function in the DNN model
$$\boldsymbol{\theta}^{*} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\left\{\frac{1}{2M}\sum_{m=1}^{M}\left(\widehat{v}(n,m) - \overline{V}^{x}(S^x(n,m)|\theta)\right)^2 + \frac{\eta}{2M}\left(\sum_{f=1}^{|F|}\sum_{d=1}^{|D|}\left(\theta_{f,d}^{(0)}\right)^2 + \sum_{g=1}^{G-1}\sum_{d=1}^{|D|}\sum_{d=1}^{|D|}\left(\theta_{d,d}^{(g)}\right)^2 + \sum_{d=1}^{|D|}\left(\theta_{d}^{(G)}\right)^2\right)\right\}$$
22. Update the value of the approximation parameters using step size α_n using $\Theta^{(i)} = (1-\alpha_n) \times \Theta^{(i)} + \alpha_n \times \Theta^{*(i)}, \forall i = 1,2,3$
23. **End for**
24. Return the approximate value function $\overline{V}^{x}(\bullet|\boldsymbol{\theta})$ for real-time MCI decision making

$$L(\boldsymbol{\theta}) = \frac{1}{2M}\sum_{m=1}^{M}\left(\widehat{v}(n,m) - \overline{V}^{x}(S^{x}(n,m)|\theta)\right)^{2} + \frac{\eta}{2M}\left(\sum_{f=1}^{|F|}\sum_{d=1}^{|D|}\left(\theta_{f,d}^{(0)}\right)^{2} + \sum_{g=1}^{G-1} \times \sum_{d=1}^{|D|}\sum_{d=1}^{|D|}\left(\theta_{d,d}^{(g)}\right)^{2} + \sum_{d=1}^{|D|}\left(\theta_{d}^{(G)}\right)^{2}\right) \tag{28}$$

$$\boldsymbol{\theta}^{*} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\{L(\boldsymbol{\theta})\} \tag{29}$$

Once $\boldsymbol{\theta}^{*}$ is computed, we apply the polynomial step-size rule, as defined in Eq. (30), to smoothly update the weights $\left(\Theta^{*(1)}, \Theta^{*(2)}, \Theta^{*(3)}\right)$ in the input, hidden, and output layers by blending them with the previous estimates $\left(\Theta^{(1)}, \Theta^{(2)}, \Theta^{(3)}\right)$, respectively. We then update $\boldsymbol{\theta}$ by Eq. (31).

$$\alpha_{n} = \frac{1}{n^{\rho}} \tag{30}$$

$$\Theta^{(i)} = (1-\alpha_{n}) \times \Theta^{(i)} + \alpha_{n} \times \Theta^{*(i)}, \forall i = 1,2,3 \tag{31}$$

Once $\boldsymbol{\theta}$ is updated using Equation (31), one iteration of the policy improvement phase is completed. The updated $\boldsymbol{\theta}$, which represents new updated fixed policy, is then used to perform the next policy evaluation phase. After repeating the policy evaluation and policy improvement phases for N iterations, the DNN-API algorithm produces a post-decision state approximate value function $\overline{V}^{x}(\cdot|\boldsymbol{\theta})$ that can be used for real-time decision making in MCI scenarios. In the DNN-API algorithm, the parameters N, M, η, ρ, $|D|$ and G, are tunable, where N is the number of iterations of the policy improvement phase, M is the number of iterations of the policy evaluation phase, η is the regularization term, ρ is the polynomial step-size parameter, $|D|$ is the number of activation nodes in each hidden layer and G is the number of hidden layers.

To balance exploration and exploitation during the offline simulation-based learning process, an ε-greedy learning policy is incorporated into the proposed DNN-API algorithm during the policy evaluation phase. Under this rule, the algorithm selects the greedy action induced by the current DNN-based post-decision-state value-function approximation with probability $1-\varepsilon_{n}$, and randomly selects a feasible exploratory action from the current feasible action set $\chi(S_{t})$ with probability ε_{n}.This mechanism is intended to diversify the simulated trajectories, expand the explored regions of the feasible action space, and reduce the risk of repeatedly following a premature greedy policy when the value-function approximation is still inaccurate in early iterations. The exploration probability is gradually reduced across policy improvement iterations according to $\varepsilon_{n} = \max\{\varepsilon^{\min}, \varepsilon_{n} \times \psi^{n}\}$, where ε_{0} is the initial exploration probability, $\varepsilon^{\min}$ is the minimum exploration probability, and $\psi \in (0,1)$ is the decay factor. As a result, the algorithm places greater emphasis on exploration in the early learning stage and gradually shifts toward exploitation as the DNN-based value-function approximation becomes more reliable. Because the proposed SDADCD problem has a high-dimensional and uncountable state space and a large combinatorial action space, the ε-greedy rule should be regarded as an exploratory sampling mechanism rather than a guarantee of exhaustive visits to all state-action pairs. After offline training, the ε-greedy exploration mechanism is no longer used in online deployment. Instead, the learned policy is implemented as a deterministic greedy policy with respect to the trained DNN-based post-decision-state value-function approximation, thereby providing stable, interpretable, and operationally reliable ambulance dispatch and casualty distribution decisions.

Building on the DNN-API algorithm described above, we further clarify the theoretical foundations of the proposed simulation-based DNN-API framework. The proposed algorithm is developed upon the well-established foundations of neuro-dynamic programming, approximate dynamic programming, simulation-based policy iteration, and value-function approximation (Bertsekas and Tsitsiklis, 1996; Powell, 2007; Bertsekas, 2011, 2019). These methods are particularly suitable for large-scale stochastic dynamic systems in which exact dynamic programming is computationally infeasible due to high-dimensional or continuous state spaces, combinatorial action spaces, long decision horizons, and analytically intractable transition probabilities (Powell, 2007; Chang et al., 2013; Bertsekas, 2019). This setting is consistent with the proposed SDADCD problem, where the MDP has a high-dimensional and uncountable state space and the state transition probabilities cannot be accurately expressed in closed form. Accordingly, the proposed DNN-API algorithm follows the classical ADP/API principle of replacing the exact value function with an approximate value function, using simulation-generated samples to estimate Bellman value updates, and iteratively improving the decision policy (Bertsekas and Tsitsiklis, 1996; Powell, 2007; Bertsekas, 2011).

The methodological rationale of the proposed framework is further supported by several related streams of literature. Bertsekas and Tsitsiklis (1996) establish the foundation of neuro-dynamic programming by integrating dynamic programming, simulation, stochastic approximation, and neural-network-based function approximation for complex sequential decision-making problems. Powell (2007) provides a comprehensive ADP framework for addressing the curses of dimensionality in large-scale stochastic dynamic resource allocation problems. Tsitsiklis and Van Roy (1997) analyze the convergence behavior of temporal-difference learning with function approximation, providing an important theoretical basis for approximate policy evaluation. In addition, Lagoudakis and Parr (2003), Munos (2003), and Munos and Szepesvári (2008) study approximate policy iteration, least-squares policy iteration, and fitted value iteration, showing how approximation errors, sampling errors, and policy evaluation errors propagate through approximate Bellman updates and affect the quality of the resulting policy. These studies provide relevant theoretical support for understanding the convergence behavior, statistical error, variance-related effects, and finite-time performance of value-function approximation methods.

The use of a deep neural network in the proposed DNN-API algorithm is also consistent with the literature on fitted Q-iteration, neural fitted Q-iteration, and deep value-function approximation. Antos et al. (2007, 2008) provide finite-time performance analyses for fitted value-based learning methods, while Riedmiller (2005) demonstrates the use of neural networks for data-efficient fitted Q-iteration. More recent studies further provide theoretical support for deep value-function approximation. For example, Fan et al. (2020) analyze the algorithmic and statistical convergence behavior of deep Q-learning and fitted Q-iteration with neural networks, showing how algorithmic error and statistical error jointly influence convergence. Gaur et al. (2023) further establish sample-complexity guarantees for fitted Q-iteration with two-layer ReLU neural networks under their modeling assumptions. These studies indicate that, under appropriate assumptions on the function class, sampling process, and approximation error, deep value-function approximation can be theoretically analyzed in terms of convergence behavior, statistical error, and sample complexity.

Therefore, the proposed simulation-based DNN-API algorithm is theoretically grounded in the ADP, neuro-dynamic programming, approximate policy iteration, fitted value iteration, fitted Q-iteration, neural fitted Q-iteration, and deep value-function approximation literature. Rather than treating the proposed method as a purely empirical algorithm, this study positions it as a structured extension of simulation-based ADP/API that uses hybrid agent-based and discrete-event simulation to generate transition samples and deep neural networks to approximate the post-decision-state value function for large-scale stochastic dynamic ambulance dispatch and casualty distribution decisions under uncertainty.

5. Hybrid agent-based and discrete event simulation

Due to the complexity of the model and the consideration of various uncertain factors, such as ambulance travel time, patient arrival time at

CCPs, and firefighter service time, the state transition probability matrix becomes uncountable and high-dimensional. Therefore, instead of using a traditional probability matrix, this study employs a mass casualty incident simulation system. The simulation system assumes that patients arrive at the CCP and complete the triage process. If a patient is classified as an immediate- or delayed-class case, the patient must be transported to a hospital by ambulance. As shown by the black route in Fig. 6, the ambulance travels from the CCP to transport the immediate-class or delayed-class patient to the hospital, and this sequence of processes can be divided into multiple discrete events for simulation. In contrast, the green route in Fig. 6 represents the movement of minor-class patients in the system. Since minor-class patients are not severely injured, they do not require immediate ambulance transportation; however, they are able to act independently and may choose to go to a shelter or directly to a hospital based on their own judgment. To simulate this autonomous behavior, agents with individual decision-making logic are introduced into the simulation system. Therefore, this study adopts a hybrid agent-based and discrete-event simulation framework, where the term "hybrid" refers to the integration of discrete-event simulation and agent-based simulation within a single simulation framework. The discrete-event simulation component is used to model the event-driven processes of the system, including patient triage completion, ambulance arrival at a CCP, and ambulance arrival at a hospital. Meanwhile, the agent-based simulation component is used to represent the autonomous behavior of minor-class patients, who may independently choose to travel to a shelter or directly to a hospital based on their own judgment. Thus, the hybrid simulation framework enables us to capture both the operational dynamics of ambulance dispatch and patient transportation and the individual-level behavioral dynamics of minor-class patients.

5.1. Discrete event simulation system

The discrete event simulation system divides the system into different events. It adds these events to an event list in the order of occurrence and then executes them sequentially. The first event in the list is executed first to effectively simulate the passage of time. Next, the system state is updated according to the result of the executed event. The execution of an event may also generate a new event, which will be inserted into the event list at the appropriate chronological position. After an event is executed, the next event in the event list is executed this process repeats until the event list is empty. Fig. 7 is a flow chart illustrating the execution of different events based on the time-advancement mechanism of discrete event simulation in a mass casualty incident system. Specifically, the event of a patient finishing triage, the event of an ambulance arriving at a CCP, and the event of an ambulance arriving at a hospital are inserted into the event list according to their order of occurrence. These events are then executed sequentially through discrete time advancement until the event list is empty. The definition of each event type in the mass casualty incident simulation model will be explained below. There are three different types of events in total:

1. Patient arrival event at a CCP

When a patient is rescued in the disaster area, the system first determines the patient's type and assigns them to the nearest CCP. It then checks if the distance to the assigned CCP is greater than the set threshold and determines if the patient is classified as immediate or delayed. If these conditions are met, the travel time of the patient from the disaster area to the CCP is calculated. Then, the patient proceeds to the CCP for inspection, and, depending on the decision, may wait for an available ambulance. Finally, an event representing the ambulance's arrival at the CCP (if needed) is added to the event list.

2. Ambulance arrives at CCP

An ambulance is dispatched from the fire station to the designated CCP according to the dispatching decision. The travel time is calculated to estimate the ambulance's arrival time at the CCP. Ambulance will travel between hospitals and CCPs, meaning this event may occur repeatedly. When this event occurs, the number of available ambulances at the specific CCP increases by one.

3. Ambulance arrives at hospital

Upon arrival at a hospital, the number of available beds in that hospital is reduced by one. Then, the ambulance returns to the CCP selected by the dispatching decision. The travel time from the hospital to the CCP is calculated, and an event representing the ambulance's arrival at the CCP is added to the event list.

5.2. Agent-based simulation for autonomous behavior of minor-class patients

After a disaster occurs, the patients are classified into four levels based on their condition: minor, delayed, immediate, and deceased. Minor-class patients have relatively minor injuries and do not require ambulance transportation; they can leave the CCP directly. Depending on their current situation and destination, minor-class patients can choose their own path and direction, as shown in Fig. 8. To simulate the post-disaster evacuation of minor-class patients on foot and consider the randomness in path selection, this study constructs an agent-based simulation system. This system captures the individual activities of each agent, allowing us to simulate the actions taken by patients based on their current conditions. It takes into account the potential damage, congestion, or other restrictions on routes that may occur after a disaster.

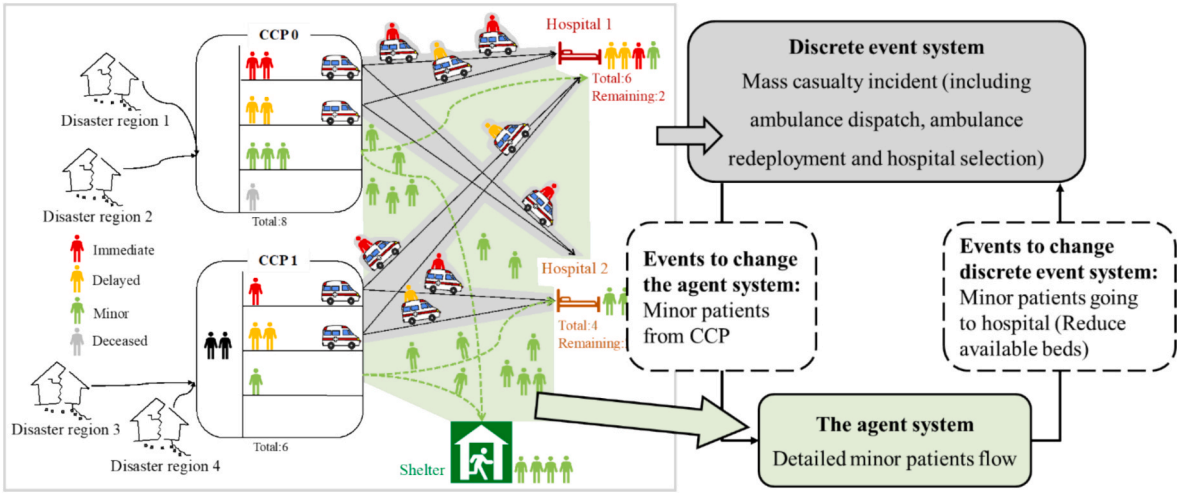


Fig. 6. Hybrid agent-based and discrete-event simulation.

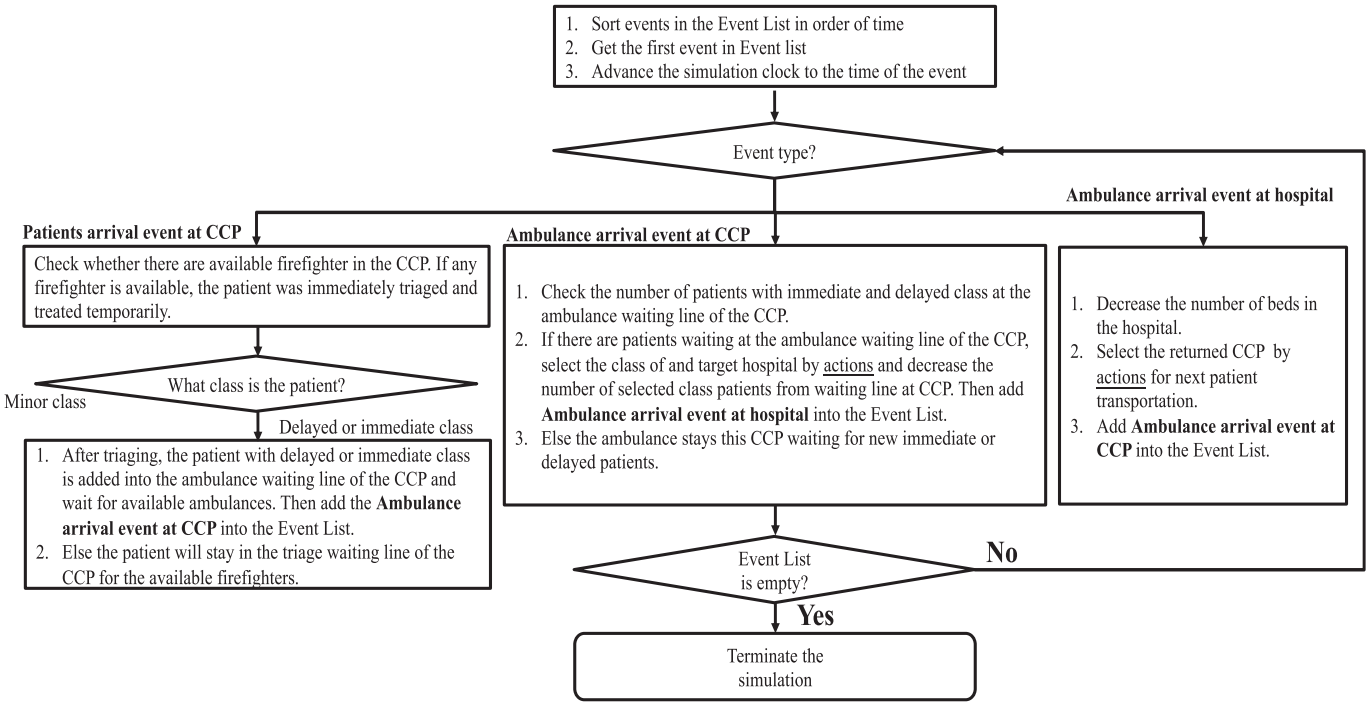


Fig. 7. Flowchart of the discrete-event simulation for the operational dynamics of ambulance dispatch and patient transportation in MCI.

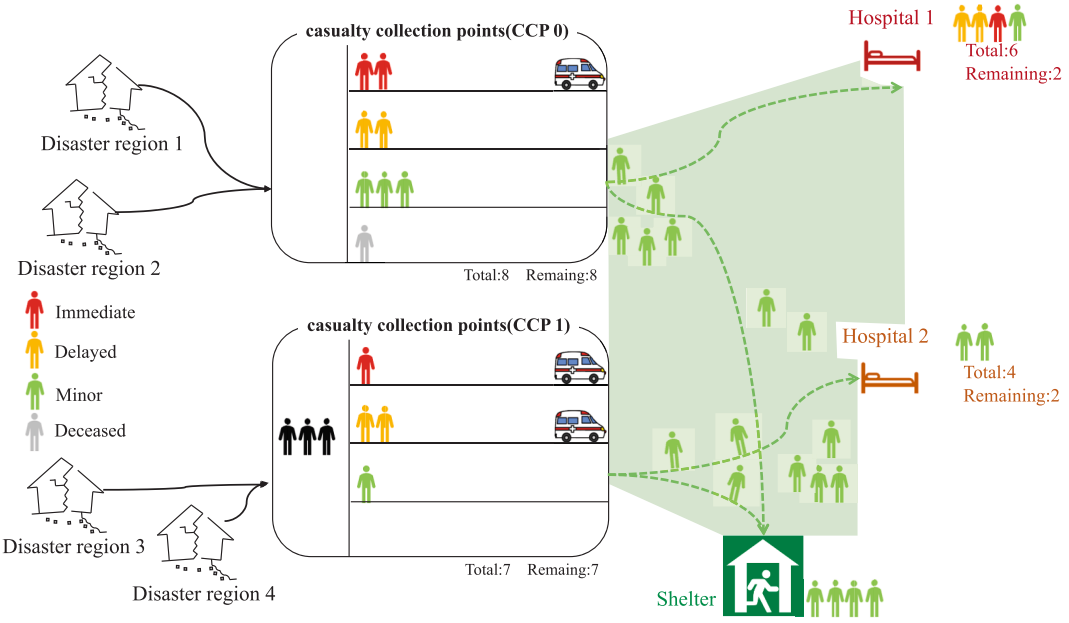


Fig. 8. Agent-based simulation for the autonomous behavior of minor-class patients.

The agent-based simulation system initially divides the road network into different grids, with intersections serving as division points. This subdivision helps to accurately simulate the paths of the patients by fine-graining the road network. The process flowchart is shown in Fig. 9. After a patient is classified as having a minor injury, a decision is made regarding whether the patient should be directed to a hospital or a shelter as their destination. Let P_h represent the probability of the minor patient going to the hospital, and P_s represent the probability of the minor patient going to the shelter, where $P_h + P_s = 1$. If the patient reaches a shelter, the simulation for that agent ends. If the patient decides to go to a hospital for treatment, the transmission model is used to simulate their path. The transmission model consists of an intra-grid transmission model and an inter-grid transmission model, which work together to achieve the simulation effect. The definitions of these two transmission models will be described in detail below. Upon arrival at the hospital, the system checks whether the hospital still has sufficient capacity to accommodate the patient. If there is enough capacity, the system's state will be updated accordingly, with the available number of beds decreasing. If there is insufficient capacity, the next destination path for the minor patient needs to be simulated again.

- **The intra-grid transmission model**

The intra-grid transmission model primarily updates the movement

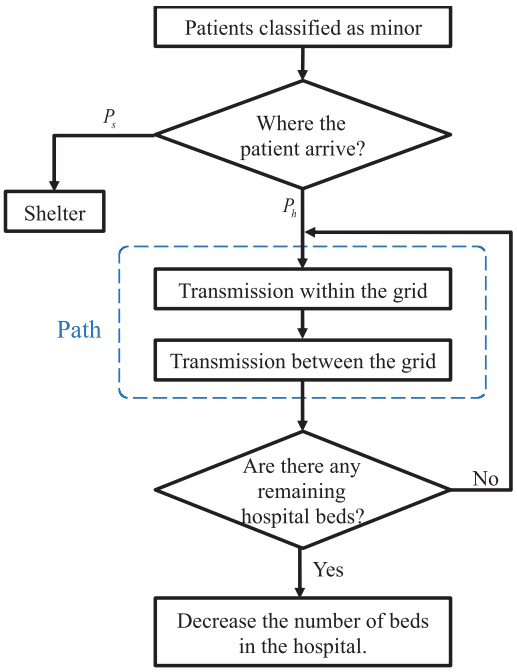


Fig. 9. Logic flow of autonomous movement of minor-class patients through the road network.

status of minor-class patients within each grid. This includes the flow direction of patients, whether they have traversed the grid, and their readiness to move to the next grid. The model randomly selects a destination based on travel time and moves the patient along the route to the grid exit. Let $T_c^{1st}(t)$ represent the shortest travel time for a minor patient to reach a hospital from grid c. Let $P_c^{T^{1st}}(t)$ represent the probability that a minor patient chooses the nearest hospital as the destination, which offers the shortest travel time from grid c (Equation (32)). When there is a significant difference in travel times between the routes, it increases the likelihood that the patients will choose the destination with the shortest travel time.

$$P_c^{T^{1st}}(t) = \frac{\frac{1}{T_c^{1st}(t)}}{\sum_{k \in N_H} \frac{1}{T_c^k(t)}} \tag{32}$$

- **The inter-grid transmission model**

The inter-grid transmission model is responsible for determining the movement direction of patients at intersections. Based on the shortest travel time to reach their destination in the adjacent grid, the model randomly assigns patients to the next grid, thereby simulating their choice behavior. When patients enter a grid, they have a predefined shortest path. However, due to uncertain factors such as traffic flow, the travel time along the initially shortest path may vary. Therefore, when patients leave the grid and reach an intersection, there is a mechanism for selecting the direction to move in. This is done by calculating the travel time required to reach the destination in the available adjacent grids and using it as a basis for choosing the desired next grid. Let $P_{c\to c'}(t)$ represent the probability that a patient in grid c chooses to move to the adjacent grid c' at time t (Equation (33)). Patients use this probability to select their next grid to move to.

$$P_{c\to c'}(t) = \frac{\frac{1}{T_{c'}^{1st}(t)}}{\sum_{u \in nei_c} \frac{1}{T_u^{1st}(t)}} \tag{33}$$

Fig. 10 depicts an illustrative example of the agent-based simulation system for minor-class patients. Through this simulation system, we can gain a better understanding and analyze the behavior of minor-class patients after a disaster. The detailed steps of the agent-based simulation system are described below:

Step 1. The minor patient chooses a destination according to the probability: P_h represents the probability that the minor patient goes to a

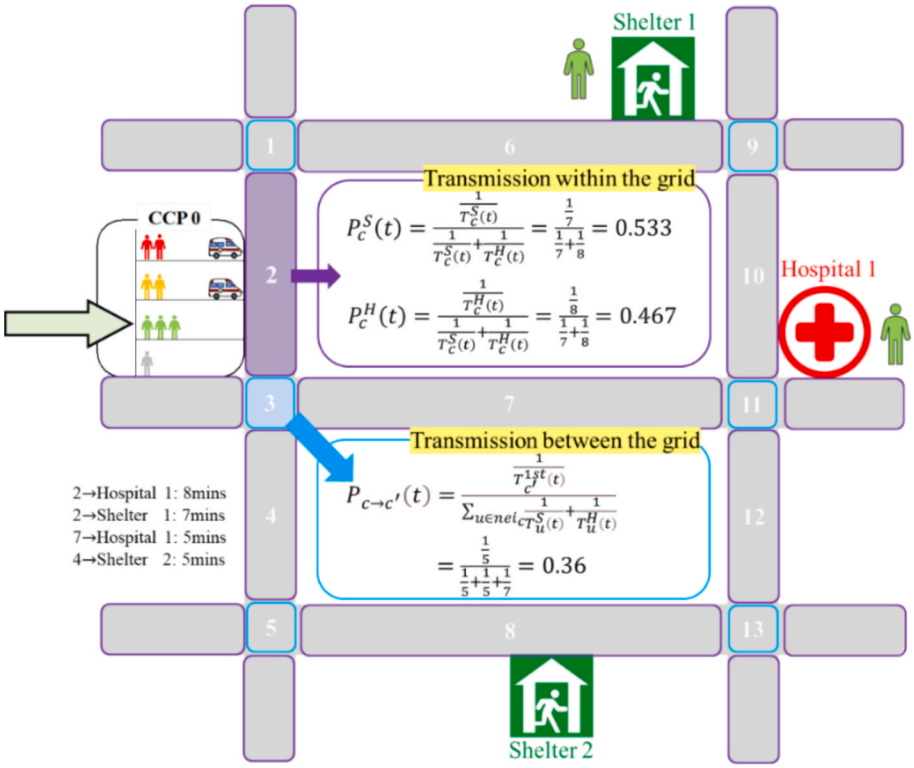


Fig. 10. An illustrative example of the agent-based simulation system operation for minor-class patients.

hospital; P_s represents the probability that the minor patient goes to a shelter.

Step 2. The minor patient chooses the hospital to go to according to the travel time. The transmission model is used to simulate their path. The transmission model consists of an intra-grid transmission model and an inter-grid transmission model. Taking the example in the Fig. 10, when the patient is in grid 2, it would take 8 min to reach Hospital 1 and 11 min to reach Hospital 2. Therefore, according to Equation (35), there is a probability of 0.579 for the minor patient to go to Hospital 1 and a probability of 0.421 to go to Hospital 2 for medical treatment.

Step 3. Minor-class patients arrive at a junction and reselect their route based on probabilities. $P_{c\to c'}(t)$ represents the probability of minor-class patients choosing the next grid to move to. Taking the example in the figure, when the patient is in grid 3, the travel time of the shortest path may change due to uncertain factors such as traffic flow, which can also affect the direction of patient movement. The patient can choose to enter an adjacent grid, i.e., grids 2, 4, or 7. The travel times required to reach their nearest destinations from these adjacent grids are 8, 8, and 5 min respectively. According to Equation (36), there is a probability of 0.278 for the minor patient to go to grid 2, a probability of 0.278 to go to grid 4, and a probability of 0.444 to go to grid 7.

Step 4. When a minor patient reaches a hospital, the system checks if the hospital has sufficient capacity to treat them. If it does, the hospital is recorded, the available beds are reduced, and the simulation for that patient ends. If the hospital does not have enough capacity, the simulation returns to Step 1 to simulate the next destination for the patient.

6. Computational study

To evaluate the computational performance of our proposed DNN-API algorithm, as well as to provide managerial insights for MCI decision-makers based on sensitivity analysis, we conduct a series of computational experiments based on a potential magnitude 6.5 earthquake scenario in the Greater Taipei area in collaboration with the National Science and Technology Center for Disaster Reduction (NCDR) in Taiwan. Section 6.1 explains the details of potential seismic scenarios and presents the optimal solution/policy for ambulance dispatch and casualty distribution. Section 6.2 presents several experiments to demonstrate the efficiency and efficacy of the proposed DNN-API algorithm. Section 6.3 conducts sensitivity analysis on five parameters, namely, the arrival rate of patients, the maximum number of available ambulances, the number of available hospital beds, the probability of minor-class patients going to the hospital and different road damage levels. All numerical experiments were performed on a computer equipped with an Intel(R) Core(TM) i7-8665U 2.1 GHz and 32 GB RAM. We used the SimPy toolbox (Matloff, 2009) to build high-resolution hybrid agent-based and discrete event simulations, and we used the Python language to implement the DNN-API algorithm.

6.1. Case study on a potential earthquake in Taiwan

One of the goals of this study is to establish a mass casualty incident simulation model and to develop a DNN-API approach to identify optimal ambulance dispatch and casualty distribution strategies within this simulation system. We set up scenarios for simulation analysis and optimal decision-making to provide responders with decision-making reference for disaster response. After discussions with the NCDR center, it was found that earthquake damage tends to be greater in urban areas. Therefore, this study will focus on major urban areas in Taiwan, specifically the Greater Taipei Area, which includes Taipei City and New Taipei City. Within the next fifty years, the Shanjiao Fault, located in the Greater Taipei metropolitan area, has the highest probability of experiencing a direct earthquake with a magnitude of 6.5 or greater. The Taiwan Earthquake Model Probabilistic Earthquake Risk Assessment (TEM PSHA2015) report also pointed out that there is a 4% probability of a magnitude 7.0 earthquake occurring on the Shanjiao Fault in the next 30 years, suggesting the non-negligible level of seismic hazard. Therefore, this study will establish a large-scale fault dislocation earthquake simulation scenario on the Shanjiao Fault. We also collect the following five relevant datasets—road network data, road damage probability, estimated casualties, fire station data, and hospital data—from the Taiwan Earthquake Impact Research and Information Application (TERIA) platform (Ke et al., 2016) of the NCDR center to construct a hybrid agent-based and discrete event simulation model. In Table 3, we summarize the values, probability distributions, and data sources of the relevant parameters for this case, and describe the quantity and geographical distribution of disaster regions, hospitals, fire stations, and casualty collection points (CCPs) in the Greater Taipei area.

We use Daan District in Taipei City as a small-scale case study, followed by the entire Greater Taipei metropolitan area as a large-scale case study. According to the earthquake simulation predictions from the NCDR, if a magnitude 6.5 earthquake occurs along the Shanjiao Fault, it is estimated that Daan District would have 525 casualties, including 249 minor injuries, 138 delayed-class patients, 79 immediate-class patients, and 59 fatalities. In the Greater Taipei metropolitan area, the number of casualties is estimated to reach as high as 12,644, including 5,831 minor injuries, 3,373 delayed-class patients, 1,944

Table 3
Relevant parameter values for the case study.

Parameters		Value, distribution, or source
Basic parameters of earthquake scenario	Number of disaster regions	Small case: 25, Large case: 229 (from TERIA platform)
	Number of hospitals	Small case: 4, Large case: 20 (from TERIA platform)
	Number of fire stations	Small case: 4, Large case: 46 (from TERIA platform)
	Number of CCP locations	Small case: 4, Large case: 16 (from TERIA platform)
	Number of total casualties	Small case: 466 (minor-class: 249, delayed-class: 138, immediate-class: 79) Large case: 3786 (minor-class: 2015, delayed-class: 1129, immediate-class: 642)
	Number of total beds	Small case: 240, Large case: 2400
	Maximum available ambulances	Small case: 5, Large case: 88
	Maximum available firefighters	Small case: 5, Large case: 92
	The rate at which patients arrive at the disaster area	Exponential distribution(λ = 10) (10 patients per hour)
	Travel speed (m/sec) of casualties from the disaster regions to the CCP locations	Uniform distribution (2, 5)
	Triage service time (sec) in CCP locations	Triangular distribution (60, 120, 180)
Parameters of travel time scenario generation	Road link damage probabilities ($p_i(x_i)$)	Based on the TERIA platform, ranging from 0% to 100%
	Threshold value ($\widehat{p}$)	0.7
	Complete road network in Taipei	Based on the TERIA platform
	The traffic jam density of route link i (D_i)	Main road: 560 (vehicles/km) Secondary road: 280 (vehicles/km)
	The free-flow speed (v_f)	18(m/sec)
	The jam speed under traffic congestion (v_{jam})	1(m/sec)
	The parameters of speed-density function (m,n)	$m = 1.23, n = 2.0$
	The traffic density of road link (d_i)	$d_i = D_i \times \widetilde{a}_i$, $\widetilde{a}_i$ Beta(3,7)
	Stochastic coefficient of traffic jam density of route link i considering road link damage probability $p_i(x_i)$ after the occurrence of the disaster ($\widetilde{\beta}_i$)	$\widetilde{\beta}_i$ Beta$(10 - 10 \times p_i(x_i), p_i(x_i))$

immediate-class patients, and 1,496 fatalities. After the earthquake, government units in the Greater Taipei metropolitan area will rapidly establish casualty collection points (CCPs). If the distance from the disaster area to the nearest hospital is less than the set threshold, the patients in the disaster area will be taken care of by nearby hospitals and will not be assigned to open CCPs. These patients will be gradually rescued from the disaster area and transferred to the nearby hospitals, reducing available beds. Otherwise, if the distance from the disaster area to the nearest hospital is greater than the set threshold, the disaster area will be assigned to the nearest CCP, and the patients will need to go through the CCP, where a decision will be made regarding whether they need to go to the hospital. In the situational road network of Daan District, there are 5 CCPs, 4 fire stations, and 4 hospitals as facility locations. In the situational road network of Taipei City, there are 16 CCPs, 46 fire stations, and 20 hospitals as facility locations. Fig. 11 presents the spatial distribution and number of key facility nodes—including disaster regions, hospitals, fire stations, and CCPs—in both the small-scale scenario (Daan District) and the large-scale scenario (Greater Taipei area). Additionally, it shows the probabilistic damage levels of road segments in the road network resulting from a magnitude 6.5 earthquake.

To consider the stochastic and dynamic nature of the MCI system in the high-resolution discrete event simulation, several input parameters, including patient inter-arrivals time, triage service time, and transportation time between different facilities, are modeled according to specific probability distributions. Based on Tseng et al. (2018), the casualty inter-arrival time (sec) rescued from the disaster area follows an exponential distribution with parameters $\lambda = 10$; the triage service time (sec) of each casualty follows a triangular distribution with parameters (60, 120, 180); and the casualty travel speed (m/sec) from disaster area to CCP location follows a uniform distribution between 2 and 5. Moreover, to incorporate the road vulnerabilities caused by the earthquake into our model, the travel time of each road segment is inflated and calculated based on the road damage probabilities generated by the TERIA platform (Chang et al., 2025). We incorporate the road segment damage probabilities to modify the speed-density function using the traffic jam density model (Jayakrishnan et al., 1994) to calculate the stochastic traffic flow speed $\widetilde{v}_i$ and the stochastic travel time $\widetilde{t}_i$ of each road segment using Equations (34) and (35). Then, based on these travel times for each road segment, we can calculate the transportation time for all possible routes between different facilities and select the route with the shortest transportation time for patient transportation.

$$\widetilde{v}_i = \begin{cases} v_{jam} & , if \widetilde{d}_i > \widetilde{D}_i \\ v_{jam} + (v_f - v_{jam}) \times \left[1 - \left(\dfrac{\widetilde{d}_i}{\widetilde{D}_i}\right)^m\right]^n & , if \widetilde{d}_i \leqslant \widetilde{D}_i \end{cases} \tag{34}$$

$$\widetilde{t}_i = \frac{L_i}{\widetilde{v}_i} \tag{35}$$

where v_f is the free-flow speed. v_{jam} is the jam speed under traffic congestion. m, n are the parameters of the speed − density function. D_i is the traffic jam density of route segment i. $\widetilde{\alpha}_i$ is a stochastic coefficient of the traffic density of route segment i right before the occurrence of the disaster. $\widetilde{\beta}_i$ is the stochastic coefficient of traffic jam density of route segment i considering road segment damage probability p_i. $\widetilde{d}_i$ is the stochastic traffic density of route segment i i right before the occurrence of the disaster, $\widetilde{d}_i = D_i \times \widetilde{\alpha}_i$. $\widetilde{D}_i$ is the stochastic traffic jam density of route segment i after the occurrence of the disaster, and $\widetilde{D}_i = D_i \times \widetilde{\beta}_i$. L_i is the length of route segment i.

6.2. Performance analysis of DNN-API algorithm

Mills et al. (2013) pointed out that the scale of a disaster and the time-dependence of survival probabilities significantly influence the

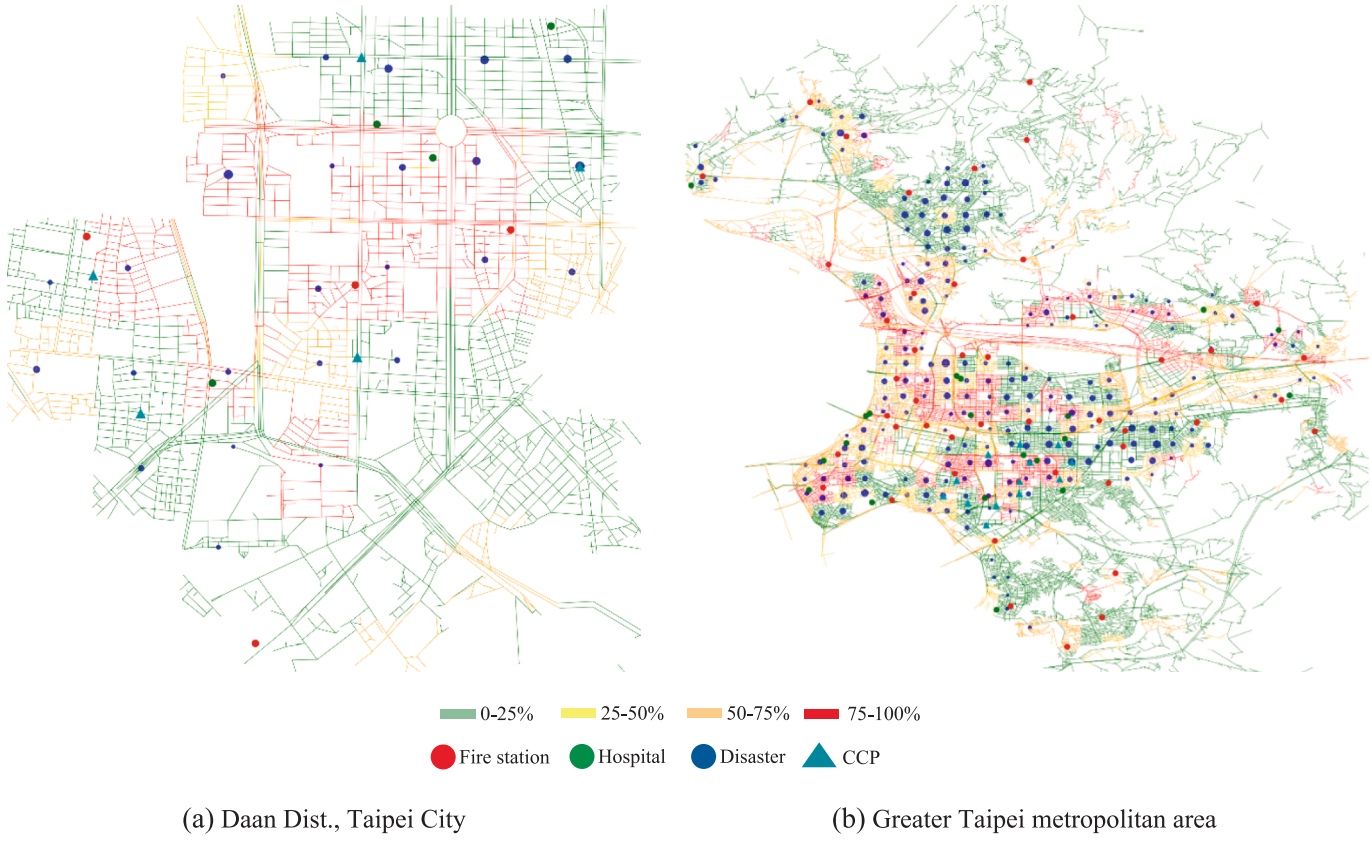


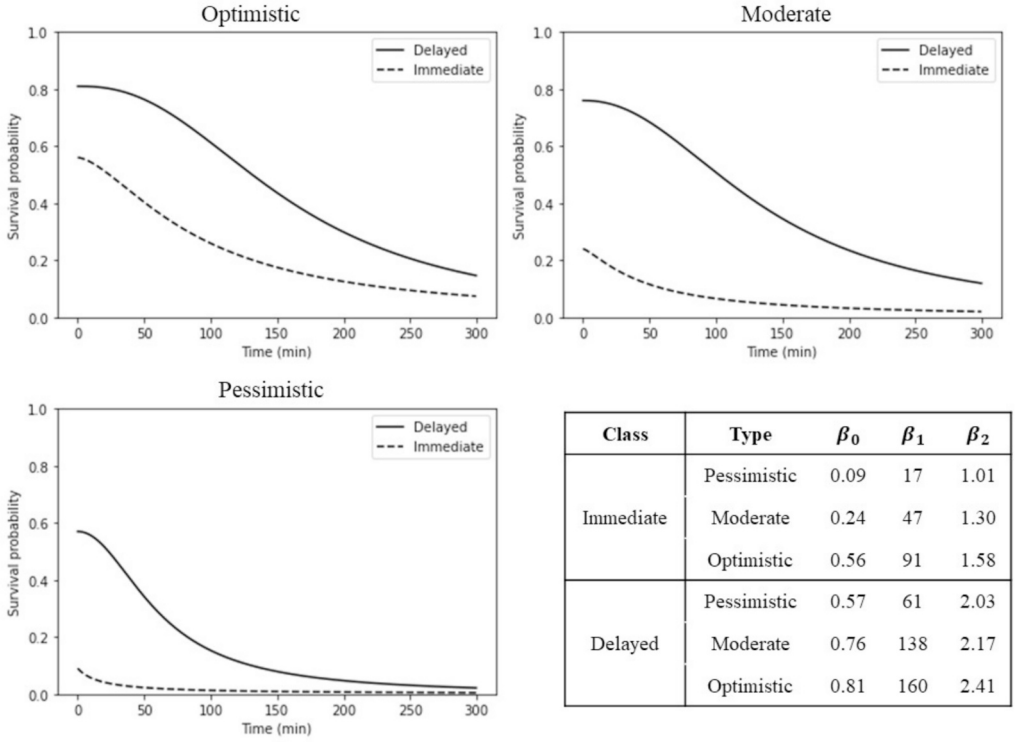
Fig. 11. A potential earthquake scenario of magnitude 6.5 caused by the Shanjiao Fault.

prioritization and scheduling of patient transportation to hospitals. Therefore, we evaluate the performance of the proposed DNN-API algorithm in making ambulance dispatch and casualty distribution decisions under varying disaster scenarios—including a small-scale scenario in Daan District and a large-scale scenario in the Greater Taipei area—as well as across different types of survival probability functions. The severity of casualties following a disaster can be characterized using different types of survival probability functions. In high-severity disasters, survival probabilities are highly time-sensitive and decline rapidly as delays increase. To capture these dynamics, we examine three representative survival probability functions—optimistic, moderate, and pessimistic—as illustrated in Fig. 12 (Mills et al., 2013; Sung and Lee, 2016). The optimistic function assumes relatively mild casualty conditions, where both immediate and delayed patients have initially high survival probabilities that decline gradually over time. In contrast, the pessimistic function reflects severe casualty conditions, where survival probabilities begin at lower levels and deteriorate sharply with even slight delays, indicating that timely treatment is critical. The moderate function represents an intermediate case between these two extremes.

In addition, to further demonstrate the effectiveness and robustness of the proposed DNN-API algorithm, we compare the differences among the DNN-API algorithm, the LSTD-based approximate policy iteration (LSTD-API), and four current practical heuristic methods under the aforementioned disaster scenarios. The LSTD-API algorithm is a commonly used approximation method within the ADP framework (Rettke et al., 2016; Jenkins et al., 2021). In the context of this study, the key difference between LSTD-API and the proposed DNN-API lies in the approximation architecture: LSTD-API adopts a linear approximation architecture to estimate the post-decision state value function. It updates the weights of the linear approximation using least-squares regression and a polynomial step-size rule. Consequently, the LSTD-API algorithm also involves three tunable parameters, namely N, M and ρ. In current MCI operations, commonly used practical heuristic methods include two types of patient prioritization strategies: (1) the Immediate-First (IF) strategy, which prioritizes immediate patients before delayed ones, and (2) the First-In-First-Out (FIFO) strategy, which transports patients in the order of their arrival, regardless of severity. For hospital selection, two common strategies are applied: (1) the Nearest-Hospital-First (NHF) strategy, which selects the hospital closest to the casualty collection point (CCP), and (2) the Cycle-Hospital (CH) strategy, which assigns patients to hospitals in a rotating manner based on their proximity to the CCP, from the nearest to the farthest. Based on the combination of these strategies, four practical heuristics can be constructed for performance comparison: IF-NHF, IF-CH, FIFO-NHF, and FIFO-CH. Since both the LSTD-API and DNN-API algorithms contain tunable parameters, we designed and conducted 3^k full factorial computational experiments to identify the optimal parameter combinations and the corresponding ambulance dispatch and casualty distribution policies for each algorithm under different disaster scenarios. Table 4 presents the parameter settings for both algorithms, with each parameter configured at three different levels. In addition, for the ε-greedy learning policy incorporated in the offline simulation-based policy evaluation phase of the DNN-API algorithm, the initial exploration probability, minimum exploration probability, and decay factor are set to $\varepsilon_0 = 0.2$, $\varepsilon^{\min} = 0.05$, and $\psi = 0.98$, respectively. Based on the optimal ambulance dispatch strategies derived from the LSTD-API and DNN-API algorithms,

Table 4
Experimental design factor levels for the algorithmic parameters of LSTD-API and DNN-API.

Algorithms	Parameters	Parameter settings
LSTD-API	N	{5, 10, 15}
	M	{100, 500, 1000}
	ρ	{0.5, 0.7, 0.9}
DNN-API	N	{5, 10, 15}
	M	{100, 500, 1000}
	η	{0.001, 0.01, 0.03}
	ρ	{0.5, 0.7, 0.9}
	$\|D\|$	{3, 5, 7}
	G	{3, 7, 10}



Class	Type	β_0	β_1	β_2
Immediate	Pessimistic	0.09	17	1.01
	Moderate	0.24	47	1.30
	Optimistic	0.56	91	1.58
Delayed	Pessimistic	0.57	61	2.03
	Moderate	0.76	138	2.17
	Optimistic	0.81	160	2.41

Fig. 12. Different types of survival probability functions (Mills et al., 2013).

as well as the four current practical heuristic strategies, we conducted 30 simulation runs for each strategy under different scenarios. We collected two performance indicators for each strategy under each simulation replication: the total survival probability of all immediate- and delayed-class patients, and the total completion time of all immediate- and delayed-class patients. Then the improvement gaps of the LSTD-API and DNN-API algorithms relative to the four practical heuristics in terms of total survival probability (TSP) and total completion time (TCT) are calculated using Equation (36) and (37) below under each simulation replication.

$$\text{Improvement gap}(\%) = \frac{\text{TSP } of \text{ API strategy}(i) - \text{TSP } of \text{ practical heuristics}(j)}{\text{TSP } of \text{ API strategy}(i)} \times 100$$
$$\forall i \in \{\text{DNN - API, LSTD - API}\},\ j \in \{\text{IF - NHF, IF - CH, FIFO - NHF, FIFO - CH}\} \tag{36}$$

$$\text{Improvement gap}(\%) = \frac{\text{TCT } of \text{ practical heuristics}(j) - \text{TCT of API strategy}(i)}{\text{TCT } of \text{ practical heuristics}(j)} \times 100$$
$$\forall i \in \{\text{DNN - API, LSTD - API}\},\ j \in \{\text{IF - NHF, IF - CH, FIFO - NHF, FIFO - CH}\} \tag{37}$$

Tables 5 and 6 summarize the mean and standard deviation of the improvement gaps in total survival probability (TSP) and total completion time (TCT) for the small-scale case in Daan District. These results compare the performance of strategies generated using the top four parameter combinations of the LSTD-API and DNN-API algorithms against those of the four practical heuristic strategies, under different survival probability functions. Similarly, Tables 7 and 8 present the corresponding statistics for the large-scale case in the Greater Taipei area, highlighting the performance differences in TSP and TCT between the LSTD-API and DNN-API algorithms and the practical heuristics across the same set of survival probability functions. From these results, we can derive several insights to illustrate the superiority of the DNN-API algorithm. The first insight is that simulation-based ADP methods, including both LSTD-API and DNN-API, consistently outperform the four practical heuristics in all earthquake disaster scenarios in terms of both total survival probability and total completion time (All improvement gaps are positive). The second insight is that, since the improvement gaps achieved by the DNN-API algorithm are consistently greater than those of the LSTD-API algorithm, DNN-API outperforms LSTD-API in both total survival probability and total completion time across all earthquake disaster scenarios. The third insight is that when the survival probability function is pessimistic, the DNN-API algorithm achieves significantly greater improvement compared to other strategies. This indicates that in scenarios where patients' survival probabilities deteriorate rapidly, the DNN-API algorithm can make more effective ambulance dispatch and casualty distribution decisions, thereby reducing the risk of death caused by treatment delays and substantially enhancing overall system performance. This result further highlights the importance and practical value of the proposed DNN-API algorithm in responding to high-severity disaster scenarios. The fourth insight is that in the large-scale case of the Greater Taipei area (see Tables 7 and 8), the improvement achieved by the DNN-API algorithm in terms of total survival probability and total completion time is more significant than in the small-scale case of Daan District (see Tables 5 and 6). This indicates that as the scale and impact of the earthquake disaster increase—such as a larger number of casualties, a wider affected area, more disaster sites, and more complex road damage—the DNN-API algorithm is capable of making more effective ambulance dispatch and casualty distribution decisions, thereby improving patient survival rates and accelerating the completion of rescue operations for all immediate-class and delayed-class casualties. The final insight is that, among the top four parameter combinations yielding the best performance of the DNN-API algorithm, the number of hidden layers is consistently set to 10. This indicates that using a deep neural network with multiple hidden layers to approximate the post-decision state value function can lead to improved algorithmic performance.

6.3. Sensitivity analysis

In this section, we perform a sensitivity analysis by varying the casualty arrival rate, the number of maximum available ambulances, the number of available hospital beds, the probability of minor-class patients seeking hospital care, and different road damage levels. All experiments are conducted based on the small-scale potential earthquake scenario in Daan District described in Section 6.1. For each scenario, we conduct 30 simulation replications to evaluate the optimal ambulance dispatch and casualty distribution strategy determined by the DNN-API algorithm. We then calculate the expected total survival probability (TSP) and total completion time (TCT) for all immediate- and delayed-class patients, and generate box plots for visual analysis. Subsequently, statistical tests are performed to determine whether there are significant differences among the different parameter levels. In addition, we further investigate how the improvement gap in total survival probability (TSP) between the DNN-API algorithm and the four current practical heuristic strategies varies across different parameter levels.

6.3.1. Influence of casualty arrival rates

Due to differences in terrain, transportation networks, and infrastructure, the difficulty of rescue operations can vary across different geographical environments. For instance, in mountainous or remote areas, roads may be rugged and inaccessible, transportation may be limited, or large-scale building collapses may block passage, all of which can affect the travel speed of rescue vehicles and the time required to transport patients from disaster sites to casualty collection points (CCPs). Therefore, this study investigates the impact of varying patient arrival rates or the time intervals between rescue arrivals at CCPs on the overall performance of the MCI system, as well as the effectiveness of the ambulance dispatch and patient distribution strategies generated by the DNN-API algorithm under different scenarios. As illustrated by the box plot trends in Fig. 13, overall rescue performance declines as the casualty arrival rate increases: total survival probability gradually decreases, while the total completion time for all immediate and delayed casualties

Table 5
Improvement gap comparison of DNN-API and LSTD-API over IF-NHF and IF-CH in the small-scale case in Daan District.

Survival probability functions	LSTD-API parameter settings			DNN-API parameter settings						Improvement gap over different policies (%)							
										LSTD-API				DNN-API			
										Immediate-first and nearest-hospital-first heuristic		Immediate-first and cycle-hospital heuristic		Immediate-first and nearest-hospital-first heuristic		Immediate-first and cycle-hospital heuristic	
	N	M	ρ	N	M	η	ρ	$\|D\|$	G	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties
Optimistic	15	1000	0.9	15	1000	0.01	0.9	7	10	12.69 ± 0.57	23.61 ±5.53	15.36 ± 1.01	56.44 ± 2.92	20.45 ±2.14	39.77 ± 2.43	26.41 ±4.22	69.67 ±5.32
	15	500	0.9	15	1000	0.001	0.9	7	10	10.95 ± 0.35	22.27 ±5.47	14.97 ±1.27	53.98 ± 2.25	19.93 ±2.11	37.88 ± 2.11	24.89 ±4.27	68.11 ±4. 32
	15	1000	0.7	15	1000	0.01	0.7	5	10	9.88 ± 0.75	19.28 ±6.88	14.31 ±1.36	50.56 ± 3.11	19.42 ±2.09	35.90 ± 3.45	23.41 ±3.99	65.23 ±4.31
	10	500	0.9	15	500	0.001	0.9	5	10	8.99 ± 0.57	17.73 ±6.37	13.88 ± 0.99	50.79 ± 3.08	18.77 ±1.89	35.34 ± 3.72	23.24 ±3.37	65.57 ±3.91
Moderate	15	1000	0.9	15	1000	0.01	0.9	7	10	11.39 ± 0.51	19.64 ± 2.09	24.54 ±1.11	50.52 ±3.41	21.65 ±1.66	39.53 ±2.14	37.36 ±5.24	65.61 ±2.15
	15	1000	0.7	15	1000	0.001	0.7	7	10	9.96 ± 0.24	9.51 ± 0.37	24.81 ± 0.36	48.79 ± 0.37	20.42 ±1.55	43.48 ±0.22	35.62 ±3.1 5	63.22 ±5.88
	10	500	0.7	15	500	0.01	0.7	7	10	8.58 ± 0.08	7.92 ±0.45	22.22 ±0.52	48.45 ±0.47	18.12 ±1.64	43.17 ±0.88	33.31 ±2.33	62.47 ±4.19
	15	500	0.5	15	500	0.001	0.5	5	10	7.14 ± 0.19	7.45 ± 0.81	20.20 ± 0.76	47.43 ± 0.74	17.64 ±1.36	38.75 ± 0.11	29.52 ±3.3 2	59.47 ±5.21
Pessimistic	15	1000	0.9	15	1000	0.01	0.9	7	10	29.54 ±4.37	14.22 ± 2.56	97.37 ±9.78	52.78 ±5.32	42.33 ±8.24	57.63 ±5.83	106.42 ±6.68	72.27 ±4.11
	10	500	0.7	15	1000	0.001	0.9	5	10	25.07 ±4.18	18.12 ± 1.76	95.44 ±9.45	48.51 ±4.36	40.76 ±6.12	52.61 ±4.71	103.87 ±4.51	71.75 ±3.69
	15	1000	0.7	15	1000	0.01	0.7	7	10	21.11 ±3.39	15.34 ±0.88	93.21 ±9.11	48.01 ±3.50	38.99 ±7.89	50.89 ±3.28	101.75 ±5.63	69.57 ±3.77
	10	1000	0.5	15	1000	0.001	0.7	5	10	20.08 ±3.11	14.23 ± 0.81	89.11 ±8.01	47.84 ±4.33	35.41 ±6.67	49.47 ±3.73	99.13 ±6.11	67.41 ±5.21

*$(\mu \pm \sigma)$:μ is mean value and σ is standard deviation.

Table 6
Improvement gap comparison of DNN-API and LSTD-API over FIFO-HNF and FIFO-CH in the small-scale case in Daan District.

Survival probability functions	LSTD-API parameter settings			DNN-API parameter settings						Improvement gap over different policies (%)					
										LSTD-API			DNN-API		
										FIFO and nearest-hospital-first heuristic	FIFO and cycle-hospital heuristic		FIFO and nearest-hospital-first heuristic	FIFO and cycle-hospital heuristic	
	N	M	ρ	N	M	η	ρ	$\lvert D\rvert$	G	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties
Optimistic	15	1000	0.9	15	1000	0.01	0.9	7	10	12.54 ± 0.22	23.63 ±0.93	17.12 ±4.61	55.58 ±2.56	21.59 ±2.33	39.7 4 ± 3.43	26.66 ±2.19	71.61 ±3.32
	15	500	0.9	15	1000	0.001	0.9	7	10	11.32 ± 0.11	22.28 ±0.97	16.71 ±2.82	54.77 ±3.12	19.81 ±2.31	37.75 ±4.43	24.27 ±2.66	68.24 ±4.41
	15	1000	0.7	15	1000	0.01	0.7	5	10	10.78 ± 0.19	19.79 ± 0.43	14.88 ±3.11	53.12 ±2.79	19.77 ±2.45	35.12 ±3.11	22.13 ±2.68	66.3 3± 3.91
	10	500	0.9	15	500	0.001	0.9	5	10	10.44 ± 0.88	18.09 ±0.50	13.76 ±3.07	52.19 ±4.78	19.45 ±1.98	35.33 ±5.28	21.39 ±1.27	65.61 ± 3.67
Moderate	15	1000	0.9	15	1000	0.01	0.9	7	10	9.21 ± 0.17	15.51 ±1.31	17.92 ±2.37	50.41 ±4.23	19.80 ±2.65	25.11 ±3.36	26.34 ±1.69	46.95 ±2.52
	15	1000	0.7	15	1000	0.001	0.7	7	10	8.10 ± 0.24	8.21 ±0.22	17.15 ±0.68	50.20 ±3.76	19.34 ±2.5 5	23.23 ±3.2 5	25.71 ±1.71	42.61 ±2.79
	10	500	0.7	15	500	0.01	0.7	7	10	7.98 ± 0.21	6.78 ±0.51	16.77 ±0.54	48.79 ±3.51	18.41 ±2.34	23.90 ±4.26	23.29 ±1.68	41.33 ±3.65
	15	500	0.5	15	500	0.001	0.5	5	10	7.78 ± 0.77	7.11 ±0.33	15.39 ±1.41	47.09 ±3.55	16.87 ±2.42	23.11 ±3.29	22.58 ±1.87	37.52 ±4.74
Pessimistic	15	1000	0.9	15	1000	0.01	0.9	7	10	20.11 ±2.19	15.21 ±1.62	64.70 ±7.41	53.88 ±5.16	32.55 ±8.46	32.86 ±4.48	73.98 ±7.31	69.17 ±3.38
	10	500	0.7	15	1000	0.001	0.9	5	10	18.99 ±2.72	14.88 ± 1.49	62.12 ±7.22	50.27 ±3.22	31.35 ±7.78	31.61 ±4.53	71.71 ±6.77	62.62 ±2.61
	15	1000	0.7	15	1000	0.01	0.7	7	10	17.17 ±3.43	13.78 ±0.27	60.91 ±6.16	49.19 ±2.97	29.87 ±4.86	29.75 ±3.21	68.60 ±7.11	60.49 ±3.63
	10	1000	0.5	15	1000	0.001	0.7	5	10	17.27 ±2.67	11.11 ± 0.68	59.51 ±5.19	48.64 ±3.07	25.48 ±3.13	28.37 ±4.28	64.36 ±6.27	57.20 ±4.31

*$(\mu \pm \sigma)$:μ is mean value and σ is standard deviation.

steadily rises. When the arrival rate is low, patients are rescued from the disaster area at a slower pace, which implies that ambulance and personnel resources are relatively sufficient and congestion within the disaster zone is minimal—conditions that contribute to higher overall survival probabilities. In contrast, a higher arrival rate means a large number of casualties are rescued in a short period of time, placing increased strain on ambulance and medical resources. This overload hinders the timely treatment of all patients, leading to a lower total survival probability. These findings indicate that variations in casualty arrival rates have a significant impact on the overall performance of the MCI system, as confirmed by ANOVA statistical tests with p value $= 0.00 \leqslant 0.05$. We also investigated the performance of various algorithms under different casualty arrival rates, with the results summarized in Table 9. As shown in Table 9, our proposed DNN-API algorithm consistently outperforms the current practical heuristic methods across all scenarios, with particularly notable improvements under lower arrival rate conditions. We believe this is because, in scenarios with lower arrival rates, rescue personnel and ambulances can reach the disaster area more quickly and provide essential medical and emergency services. Such rapid response significantly improves patient survival probabilities, and congestion within the disaster area is relatively low. Under these circumstances, the proposed DNN-API algorithm can allocate resources more flexibly to ensure that patients receive timely and effective ambulance services and are quickly transported to hospitals for treatment. Moreover, the DNN-API algorithm also demonstrates clear advantages over current practical heuristics even under high system load conditions (i.e., higher casualty arrival rates).

6.3.2. *Influence of maximum available ambulances*

Based on past experiences in large-scale casualty incidents, urban areas typically can only mobilize one-third of their total number of ambulances for rescue missions, as the remaining ambulances must remain on standby to respond to everyday medical needs. Therefore, in disaster response, additional ambulance support from neighboring areas is required. To investigate this impact, we examined the effect of varying the proportion of available ambulances on the system's performance. Taking Da'an District as an example, which has a total of 6 ambulances, we searched for the optimal solution under different numbers of available ambulances. From Fig. 14, it can be observed that as the proportion of available ambulances increases, the performance improves. Particularly when the proportion of available ambulances is below 60%, there is a relative shortage of rescue resources, leading to decreased rescue speed and efficiency, which negatively impacts the survival probability and completion time of the patients. However, as the proportion of available ambulances increases, the supply of rescue resources grows, improving the efficiency of rescue missions and the survival probability of the injured patients. It is important to note that once the proportion of available ambulances exceeds a certain threshold, the improvement in performance will gradually diminish. This is because a sufficient number of ambulances has already been reached for effective rescue operations. Increasing the number of ambulances beyond this point would not significantly enhance performance, and other disaster response methods would need to be implemented to further improve the survival probability of the patients. The one-way ANOVA statistical tests with p value $= 0.00 \leqslant 0.05$ also indicate that the objective values under different levels of maximum available ambulances have significant differences. Furthermore, as shown in Table 10, we further examined the performance of various algorithms under different levels of ambulance availability. The results indicate that the proposed DNN-API algorithm achieves significant improvements in both total survival probability and total completion time compared to current practical heuristic methods. This advantage is particularly evident when ambulance availability is low, meaning that rescue resources are relatively scarce. The DNN-API algorithm is capable of more effectively and appropriately dispatching ambulances and allocating medical resources, thereby enhancing patient survival probability. Therefore, the DNN-API algorithm developed

Table 7
. Improvement gap comparison of DNN-API and LSTD-API over IF-HNF and IF-CH in the large-scale case in the Greater Taipei area.

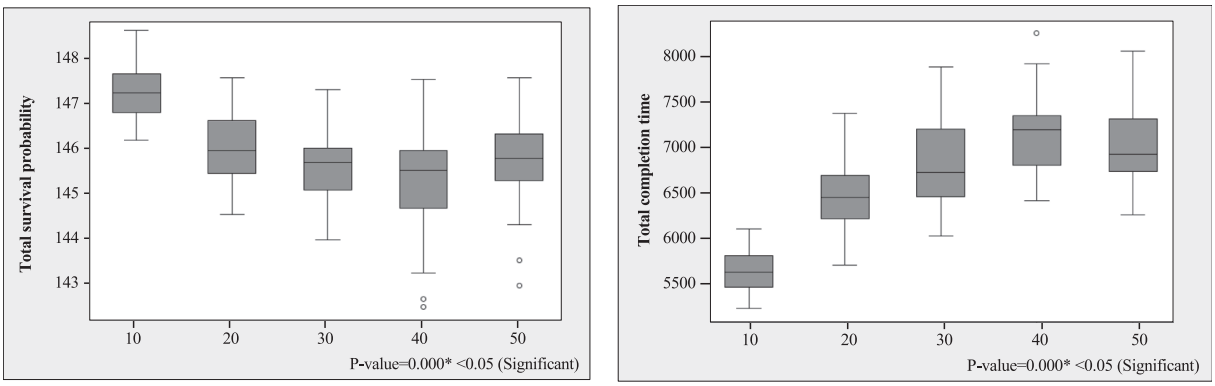
Survival probability functions	LSTD-API parameter settings			DNN-API parameter settings						Improvement gap over different policies (%)							
										LSTD-API				DNN-API			
										Immediate-first and nearest-hospital-first heuristic		Immediate-first and cycle-hospital heuristic		Immediate-first and nearest-hospital-first heuristic		Immediate-first and cycle-hospital heuristic	
	N	M	ρ	N	M	η	ρ	$\|D\|$	G	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties
Optimistic	15	1000	0.9	15	1000	0.01	0.9	7	10	16.14 ± 0.19	26.84 ± 2.09	21.99 ±2.01	62.98 ±3.38	32.77 ±2.13	47.67 ±3.88	38.58 ±3.15	84.46 ±5.38
	15	1000	0.7	15	1000	0.001	0.7	7	10	15.95 ± 0.29	24.40 ±0.15	18.25 ±1.77	59.88 ±4.53	31.19 ±1.30	46.91 ± 2.74	37.89 ±3.43	81.29 ±4.09
	15	500	0.9	15	1000	0.03	0.9	5	10	13.71 ± 0.52	21.99 ±0.67	15.29 ±1.79	58.64 ±4.56	30.58 ±2.08	43.73 ±2.12	37.53 ±4.15	79.97 ±3.76
	15	1000	0.5	15	1000	0.01	0.7	7	10	12.85 ± 0.19	20.88 ± 0.13	14.88 ±0.53	55.74 ±3.19	29.94 ±1.06	42.81 ± 3.98	34.33 ±3.42	78.62 ±3.50
Moderate	15	1000	0.9	15	1000	0.01	0.9	7	10	11.49 ± 0.22	15.26 ±1.31	20.11 ±1.21	38.16 ±3.32	30.14 ±1.96	57.61 ±2.82	47.58 ±4.75	77.25 ±2.16
	15	500	0.9	15	1000	0.01	0.7	7	10	5.78 ± 0.89	12.88 ± 1.28	20.42 ±2.21	36.05 ±3.73	29.55 ±1.25	55.83 ±2.7 5	46.21 ±5.19	76.35 ±5.40
	15	1000	0.7	15	1000	0.001	0.9	7	10	5.58 ± 0.18	10.21 ±1.65	18.77 ±1.81	28.83 ±3.31	27.92 ±1.07	54.51 ±3.11	45.20 ±4.14	75.11 ±4.36
	10	1000	0.9	15	1000	0.01	0.9	7	10	4.18 ± 0.24	9.23 ± 0.77	15.46 ±1.47	22.43 ±4.13	26.19 ±1.11	53.78 ±3.21	43.38 ±3.40	73.16 ±4.03
Pessimistic	15	1000	0.9	15	1000	0.01	0.9	7	10	36.42 ± 0.76	16.16 ±0.88	38.51 ±4.77	46.91 ±3.32	47.69 ±6.75	72.35 ±4.17	119.12 ±6.64	85.61 ±6.11
	10	1000	0.9	15	1000	0.001	0.9	7	10	17.88 ± 0.21	11.72 ±2.55	22.99 ±3.43	41.72 ±3.12	46.29 ±5.43	70.38 ±2.89	114.57 ±6.82	83.58 ±5.22
	15	1000	0.7	15	500	0.01	0.7	7	10	13.17 ± 0.44	9.85 ±0.61	16.64 ±3.39	35.88 ±2.53	45.34 ±3.17	67.81 ±3.27	111.93 ±5.27	82.18 ±3.88
	10	1000	0.7	15	1000	0.001	0.9	7	10	11.30 ± 0.28	11.23 ± 0.83	14.44 ±1.77	31.83 ±4.83	43.52 ±4.35	66.96 ±4.03	108.75 ±6.06	80.19 ±2.90

*$(\mu \pm \sigma)$:μ is mean value and σ is standard deviation.

Table 8
Improvement gap comparison of DNN-API and LSTD-API over FIFO-HNF and FIFO-CH in the large-scale case in the Greater Taipei area.

Survival probability functions	LSTD-API parameter settings			DNN-API parameter settings						Improvement gap over different policies (%)							
										LSTD-API				DNN-API			
										Immediate-first and nearest-hospital-first heuristic		Immediate-first and cycle-hospital heuristic		Immediate-first and nearest-hospital-first heuristic		Immediate-first and cycle-hospital heuristic	
	N	M	ρ	N	M	η	ρ	$\|D\|$	G	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties	Total survival probability	The completion time of delayed and immediate casualties
Optimistic	15	1000	0.9	15	1000	0.01	0.9	7	10	17.01 ± 0.82	31.44 ±1.13	22.02 ±3.11	61.11 ±4.64	28.47 ±2.55	47.4 4 ± 4.33	38.28 ±1.98	87.75 ±3.09
	15	1000	0.7	15	1000	0.001	0.7	7	10	15.33 ± 0.31	29.05 ±0.93	20.45 ±2.26	59.22 ±3.28	27.13 ±2.42	45.1 4 ± 3.07	37.27 ±2.15	86.04 ±3.37
	15	500	0.9	15	1000	0.03	0.9	5	10	14.02 ± 0.19	28.99 ±0.88	18.67 ±2.74	58.49 ±2.84	26.27 ±1.88	42.85 ±4.69	35.04 ±1.41	83.29 ±2.54
	15	1000	0.5	15	1000	0.01	0.7	7	10	11.44 ± 0.74	28.32 ±1.01	16.85 ±2.22	57.99 ±4.26	25.11 ±1.44	41.99 ±5.23	31.30 ±1.03	79.37 ±2.16
Moderate	15	1000	0.9	15	1000	0.01	0.9	7	10	23.89 ± 0.69	15.13 ±1.03	16.83 ±1.99	46.34 ±3.93	27.92 ±1.98	36.11 ±3.25	43.33 ±1.23	57.50 ±2.16
	15	500	0.9	15	1000	0.01	0.7	7	10	17.18 ± 0.17	11.26 ± 1.83	14.20 ±0.84	44.17 ±2.85	27.48 ±2.11	35.17 ±3.6 5	40.71 ±1.0 5	56.93 ±2.2 5
	15	1000	0.7	15	1000	0.001	0.9	7	10	13.11 ± 0.95	10.33 ±0.47	12.31 ±1.06	43.39 ±3.71	26.60 ±1.95	34.18 ±4.01	39.28 ±1.37	56.16 ±2.46
	10	1000	0.9	15	1000	0.01	0.9	7	10	10.19 ± 0.12	11.28 ± 0.84	10.71 ±1.13	30.37 ±3.55	23.34 ±2.25	33.70 ±3.85	35.11 ±1.79	52.99 ±3.01
Pessimistic	15	1000	0.9	15	1000	0.01	0.9	7	10	26.78 ± 0.17	14.31 ±1.03	25.81 ±2.14	50.41 ±4.71	42.09 ±5.01	48.22 ±4.12	78.22 ±5.14	93.01 ±3.67
	10	1000	0.9	15	1000	0.001	0.9	7	10	20.84 ± 0.82	11.31 ± 1.23	24.22 ±1.91	46.33 ±2.41	39.71 ±5.41	45.38 ±3.37	77.12 ±6.04	90.12 ±3.83
	15	1000	0.7	15	500	0.01	0.7	7	10	17.02 ±1.10	10.81 ±0.58	24.80 ±1.49	38.16 ±2.78	37.70 ±6.22	44.79 ±3.11	76.07 ±6.22	88.81 ±2.84
	10	1000	0.7	15	1000	0.001	0.9	7	10	14.91 ± 0.64	11.37 ± 0.72	19.21 ±0.59	32.14 ±2.45	36.29 ±5.29	44.01 ±2.51	74.17 ±5.18	88.05 ±3.53

*$(\mu \pm \sigma)$:μ is mean value and σ is standard deviation.



(a) Total survival probability (b) Expected complete delivery time

Fig. 13. Comparison of two metrics across different arrival rate levels.

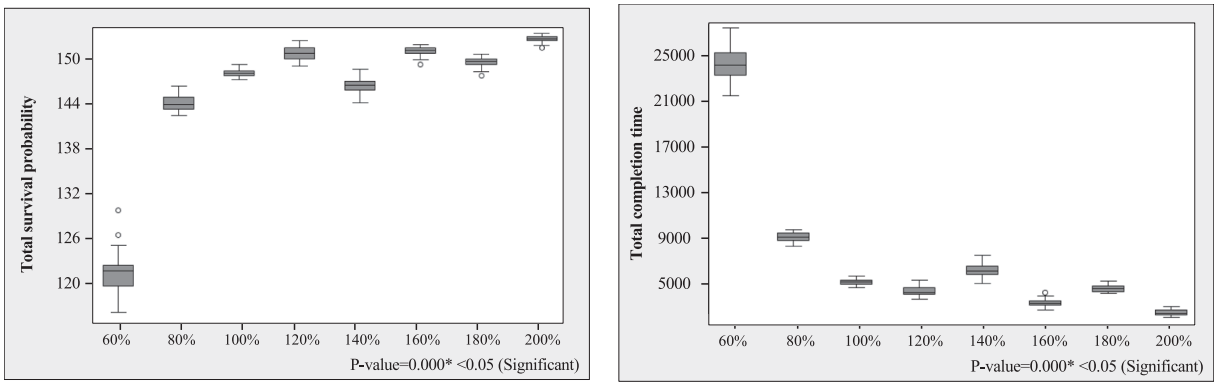
Table 9
Improvement gap of DNN-API over four practice heuristics in total survival probability of all delayed and immediate casualties under different arrival rate levels.

Different levels of arrival rates	10	20	30	40	50
IF-NHF	20.74 ± 0.61	18.42 ± 0.52	18.63 ± 0.74	18.58 ± 0.58	18.38 ± 0.77
FIFO-NHF	19.32 ± 1.02	16.90 ± 0.53	17.12 ± 0.78	16.80 ± 1.15	16.23 ± 0.45
IF-CH	26.88 ± 1.21	25.41 ± 1.26	24.66 ± 1.49	24.77 ± 1.66	23.68 ± 1.08
FIFO-CH	26.27 ± 0.88	25.94 ± 1.29	24.76 ± 1.26	24.19 ± 1.37	24.25 ± 1.26

in this study can assist frontline emergency responders in more effectively managing large-scale disaster events, ultimately reducing casualties and losses.

6.3.3. *Influence of available hospital beds*

In large-scale mass casualty incidents, the capacity and availability of hospital beds play a crucial role in patient survival and treatment outcomes. When hospital capacity is insufficient to meet post-disaster treatment demands, patients may need to be transferred to more distant hospitals. This increases travel times, delays critical treatment, and adversely affects patient survival. Expanding hospital capacity, such as increasing the number of available beds, can help mitigate these issues and improve patient survival probability. To analyze the impact of hospital capacity on performance metrics, we examined variations in performance under different numbers of available beds. The results in Fig. 15 indicate a significant overall improvement as the number of available beds gradually increases. With an increased number of available beds, hospitals can receive and treat patients more rapidly, reducing wait times and transfer delays. This leads to higher total survival probabilities and decreased total completion times. The one-way ANOVA statistical tests with p value $= 0.00 \leqslant 0.05$ also indicate that the objective values under different levels of available hospital beds have significant differences. We also compare the DNN-API algorithm

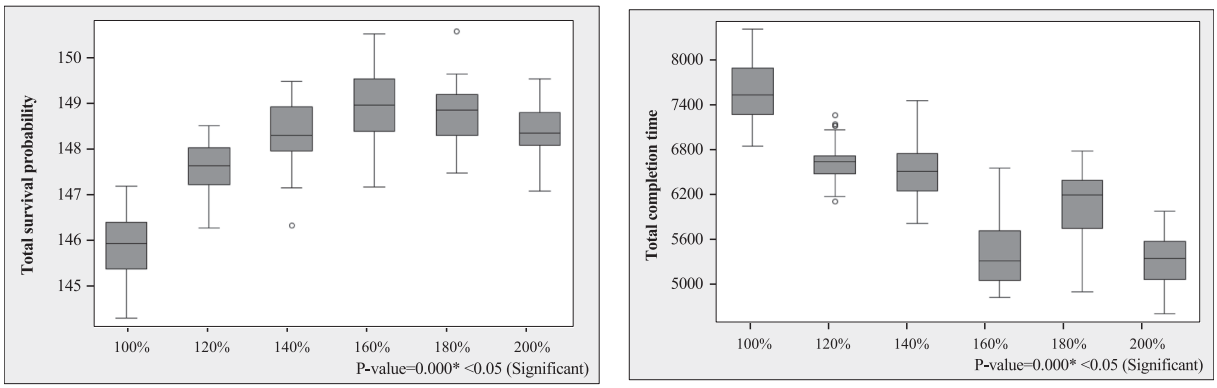


(a) Total survival probability (b) Expected complete delivery time

Fig. 14. Comparison of two metrics across different levels of available ambulance resources.

Table 10
Improvement gap of DNN-API over four practice heuristics in total survival probability for delayed and immediate casualties under different levels of available ambulance resources.

Available ambulance resources	60%	80%	100%	120%	140%	160%	180%	200%
IF-NHF	36.60 ± 4.01	37.14 ± 3.03	22.73 ± 0.62	22.18 ± 1.06	20.64 ± 0.62	18.52 ± 0.48	18.10 ± 0.71	18.90 ± 0.50
FIFO-NHF	33.73 ± 2.38	34.69 ± 2.72	23.39 ± 0.45	23.72 ± 0.57	22.59 ± 0.74	19.89 ± 0.61	19.24 ± 0.88	19.60 ± 0.61
IF-CH	53.60 ± 2.74	53.74 ± 2.53	25.87 ± 0.89	25.88 ± 1.06	25.85 ± 1.22	19.40 ± 0.69	18.98 ± 1.31	17.62 ± 0.43
FIFO-CH	57.35 ± 2.19	56.99 ± 2.47	26.80 ± 0.61	26.88 ± 1.23	26.12 ± 0.85	20.77 ± 0.58	20.45 ± 0.99	19.07 ± 0.39



(a) Total survival probability (b) Expected complete delivery time

Fig. 15. Comparison of two metrics across different levels of available hospital bed resources.

Table 11
Improvement gap of DNN-API over four practice heuristics in total survival probability for delayed and immediate casualties under varying levels of available hospital bed resources.

Available hospital beds	100%	120%	140%	160%	180%	200%
IF-NHF	20.70 ± 0.92	23.61 ± 0.99	22.22 ± 0.67	23.44 ± 0.88	23.54 ± 0.85	23.60 ± 0.51
FIFO-NHF	19.39 ± 0.76	22.19 ± 0.72	20.73 ± 0.69	22.02 ± 0.75	22.06 ± 0.77	22.05 ± 0.49
IF-CH	26.94 ± 1.25	28.82 ± 1.15	31.05 ± 1.34	31.20 ± 1.04	32.01 ± 1.25	30.27 ± 0.77
FIFO-CH	27.89 ± 0.88	29.91 ± 0.75	32.05 ± 1.21	31.95 ± 1.11	32.55 ± 1.49	31.15 ± 1.15

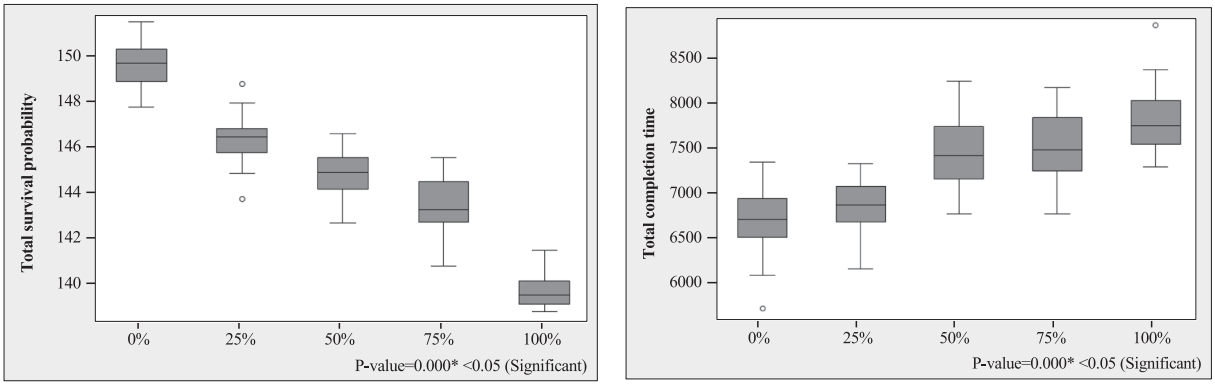
with current practical heuristics, as shown in Table 11. Across all scenarios, the DNN-API algorithm developed in this study consistently provided significant improvements in efficiency, particularly when the expansion of available beds is greater. This underscores the critical importance of increasing the number of available beds in hospitals for addressing large-scale mass casualty incidents. By expanding hospital capacity through measures such as increasing the number of beds and setting up temporary medical facilities, delays and bottlenecks can be effectively reduced, thereby improving the survival probability of the patients. However, ensuring sustainable and effective improvements in hospital capacity requires consideration of other factors, such as an adequate healthcare workforce and sufficient medical equipment and supplies. Additionally, establishing a well-coordinated mechanism is essential to ensure effective collaboration among hospitals and the efficient allocation of resources to meet demands during disaster events.

6.3.4. *Influence of probability of minor-class patients going to hospital*

Minor-class patients typically sustain relatively minor injuries and retain a certain level of independent mobility, meaning they do not require urgent transportation to hospitals by ambulance. However, they may still need some emergency services, such as wound dressing or medication application. In such cases, they may choose to go to the hospital on their own. Therefore, this experiment aims to explore the impact of varying the proportion of minor-class patients seeking self-treatment on the survival probability of immediate-class and delayed-class patients. We conducted a series of experiments, considering

Table 12
Improvement gap of DNN-API over four practice heuristics in total survival probability of all delayed and immediate casualties under different probabilities of minor-class patients going to hospitals.

Probability of minor patient going to hospital	0%	25%	50%	75%	100%
IF-NHF	21.49 ± 0.68	20.97 ± 0.75	20.98 ± 0.93	20.98 ± 1.25	18.85 ± 1.53
FIFO-NHF	20.98 ± 0.74	20.26 ±0.61	20.54 ± 0.85	20.87 ± 1.37	18.27 ± 0.67
IF-CH	29.38 ± 1.40	24.98 ± 1.42	25.81 ± 1.24	20.71 ± 1.14	22.66 ± 1.01
FIFO-CH	28.68 ± 1.39	24.24 ± 1.62	24.71 ± 1.47	19.73 ± 1.09	20.94 ± 1.99



(a) Total survival probability (b) Expected complete delivery time

Fig. 16. Comparison of two metrics across different probabilities of minor-class patients going to hospitals.

different proportions of minor-class patients seeking self-treatment, and identified the optimal solution. The experimental results in Fig. 16 reveal that the variation in the proportion of minor-class patients seeking self-treatment affects the survival probability of other patients. When the proportion of minor-class patients seeking self-treatment is higher, they occupy a portion of medical resources, thereby reducing the resources available for the treatment of immediate-class and delayed-class patients. This forces other patients to seek treatment at more distant hospitals, which lowers their survival probability. The one-way ANOVA statistical tests with p value $= 0.00 \leqslant 0.05$ also indicate that the objective values under different levels of probability of minor-class patients going to hospital have significant differences. Next, we compared the DNN-API algorithm with current practical heuristics. As shown in Table 12, compared to the current practical heuristics, our algorithm consistently yields improved total survival probabilities across different parameter settings. The DNN-API algorithm performs remarkably well in optimizing resource allocation, making it more effective at handling the challenges in post-disaster rescue operations. The algorithm offers valuable insights for enhancing response capabilities during large-scale casualty incidents. It is crucial to recognize the importance of government or rescue personnel control over self-treatment by minor-class patients after a disaster. In scenarios where minor-class patients seek self-treatment, their occupation of available hospital resources and the increased ambulance round-trip times can lead to decreased performance values. This aspect should be considered, and different dispatch decisions for ambulances should be made during post-disaster rescue operations. However, it is crucial to ensure that these patients receive timely and appropriate emergency services to prevent the exacerbation of injuries or the occurrence of other unforeseen circumstances.

6.3.5. *Influence of different road damage levels*

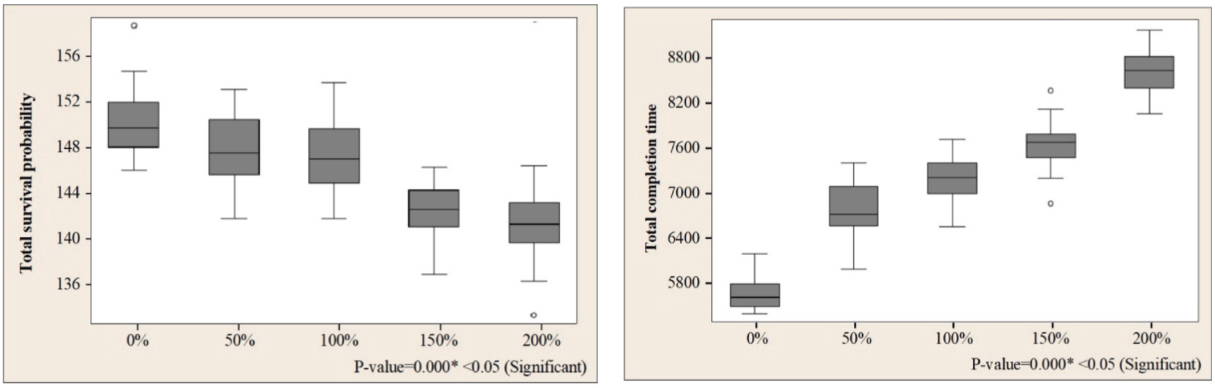
Severe earthquakes often result in substantial damage to road infrastructure, thereby significantly impeding the timely rescue and transportation of casualties. To assess the impact of road network disruption on emergency medical response, this section evaluates the effects of varying road damage levels—ranging from 0% (i.e., no damage) to 200% of the baseline values—on two critical performance metrics: the overall survival probability of casualties and the total completion time required to transport all patients to medical facilities. As illustrated in Fig. 17, increasing levels of road damage—corresponding to higher earthquake intensities—are associated with a pronounced decline in system-wide survival probability and a substantial increase in transportation completion time. The one-way ANOVA statistical tests with p value $= 0.00 \leqslant 0.05$ also indicate that the objective values under different road damage levels have significant differences. Further analysis based on Table 13 demonstrates that the performance gains achieved by the DNN-API algorithm over the IF-NHF and FIFO-NHF heuristics remain relatively consistent regardless of the extent of road damage. This finding indicates that DNN-API exhibits robust performance across a wide range of disaster scenarios. In contrast, when benchmarked against the IF-CH and FIFO-CH strategies, the relative superiority of DNN-API becomes increasingly evident as road damage intensifies, suggesting an expanding improvement gap under more adverse conditions. In summary, the results indicate that the DNN-API algorithm offers progressively greater benefits in dispatch optimization as earthquake magnitude and associated infrastructure damage increase, thereby reinforcing its value as a decision support tool for frontline emergency response operations. Additionally, the probability of road disruption emerges as a key determinant of patient transport times, highlighting the critical importance of post-disaster road restoration efforts in enhancing the efficiency and effectiveness of medical evacuation logistics.

Table 13
Improvement gap of DNN-API over four practice heuristics in total survival probability of all delayed and immediate casualties across different road damage levels.

Different levels of road damage	0%	50%	100%	150%	200%
IF-NHF	18.85 ± 1.73	20.98 ± 1.21	20.98 ± 0.69	20.97 ±1. 02	21.49 ± 0.60
FIFO-NHF	18.27 ± 0.84	20.87 ± 1.04	20.54 ± 1. 02	20.26 ±0.91	20.98 ± 0.69
IF-CH	22.66 ± 1.61	20.71 ± 1.09	25.81 ± 1.51	24.98 ± 1.21	29.38 ± 1.21
FIFO-CH	20.94 ± 1.01	19.73 ±0. 94	24.71 ± 1.35	24.24 ± 1.43	28.68 ±0.97

7. Conclusion

This study addresses the stochastic dynamic ambulance dispatch and casualty distribution problem in mass casualty incidents by formulating the problem as a Markov Decision Process. The MDP model is designed to generate optimal policies for ambulance dispatching, hospital selection, and ambulance redeployment, with the overarching objective of maximizing the expected survival probability of casualties. However, the model's high-dimensional and uncountable state space, coupled with the difficulty in precisely estimating transition probabilities, poses significant computational challenges. To overcome these limitations, we introduce a simulation-based approximate dynamic programming solution methodology that integrates a deep neural network-based approximate policy iteration (DNN-API) algorithm with a hybrid agent-based and discrete-event simulation model. The DNN-API algorithm captures the spatiotemporal complexities of MCI systems by



(a) Total survival probability (b) Expected complete delivery time

Fig. 17. Comparison of two metrics across different road damage levels.

constructing a set of basis functions to reduce the dimensionality of the state space, which are then used as input features for training a deep neural network. The network is iteratively updated using a diverse set of nonlinear function combinations, allowing for accurate approximation of the post-decision value function and improved policy learning performance. The hybrid agent-based and discrete-event simulation model complements this learning process by capturing the stochastic dynamics of the MCI logistics system at a high resolution. It accounts for key sources of uncertainty, including random casualty arrival times, variable triage and treatment durations, ambulance travel time variability, and dynamic changes in casualty conditions. Moreover, it incorporates agent-based representations of autonomous behaviors exhibited by mildly injured individuals, such as self-evacuation and self-treatment. Through the iterative interaction between the DNN-API algorithm and the hybrid agent-based and discrete-event simulation model, a near-optimal ambulance dispatch and casualty distribution strategy can be effectively generated. The integration of DNN-API with high-fidelity simulation not only enhances the realism and flexibility of the decision-making process but also provides a powerful data-driven decision support tool applicable to real-world disaster scenarios characterized by high uncertainty and dynamic change, thereby assisting in timely emergency medical response decisions.

In collaboration with the NCDR center in Taiwan, this study conducts an empirical analysis using a potential earthquake scenario (magnitude 6.5) along the Shanjiao Fault in the Greater Taipei Metropolitan Area. The results demonstrate that the proposed DNN-API algorithm significantly outperforms the traditional least-squares temporal difference-based approximate dynamic programming method (LSTD-API) and four commonly used practical heuristic strategies in terms of both total survival probability and total completion time. Especially under large-scale earthquake scenarios, including a greater number of casualties, a wider affected area, more disaster sites and more complex road damage, as well as under a pessimistic casualty survival probability function, the DNN-API algorithm achieves significantly greater improvement compared to other strategies. Through sensitivity analysis, we found that variations in casualty arrival rate, the maximum number of available ambulances, the number of available hospital beds, the probability of minor-class patients seeking hospital treatment, and road damage levels all have a significant impact on the expected total survival probability and total completion time for both immediate- and delayed-class patients. Among them, the maximum number of available ambulances is the most critical emergency medical service resource affecting total survival probability of mass casualty incidents. The number of available hospital beds, the probability of minor-class patients seeking hospital treatment, and the level of road damage are secondary yet important factors influencing both the total survival probability and the total completion time. Moreover, under various sensitivity analysis conditions, the proposed DNN-API algorithm consistently outperforms the four commonly used practical heuristic strategies by generating more efficient ambulance dispatch and casualty distribution plans. It delivers significant improvements in both total survival probability and total completion time performance metrics. In particular, under scenarios with severe road damage or extreme shortages of ambulance resources, the DNN-API algorithm demonstrates superior performance, highlighting its strong adaptability and advantage in high-stress disaster environments.

The proposed simulation-based DNN-API framework can support real-time decision-making in two potential implementation schemes. The first role is the offline simulation–online application scheme (Hong and Jiang, 2019; Jiang et al., 2020). In this role, the hybrid agent-based and discrete-event simulation model is used offline to generate representative stochastic trajectories of the MCI system, including casualty arrivals, triage processes, ambulance travel times, hospital-bed dynamics, and the self-initiated hospital visits of minor-class patients. These simulation-generated trajectories provide state-transition and reward samples for training the DNN-based post-decision-state value function approximation. Once the value function has been trained, it can be deployed online to support fast ambulance dispatch, hospital selection, and ambulance redeployment decisions whenever a new decision epoch occurs. In practical MCI response, this means that when an ambulance becomes available, the currently observed system state, such as casualty queues at CCPs, ambulance status, hospital-bed availability, and elapsed response time, can be used as input to the trained decision model. The model can then rapidly recommend the high-quality feasible ambulance dispatch and casualty distribution decisions without re-running the full simulation-based training procedure. In this implementation, "real-time" refers to rapid online decision generation using the current observed system state and the offline-trained value function, rather than online retraining during the response process.

The second role is the streaming-data-based adaptive decision-making scheme (O'Callaghan et al., 2002; Liu et al., 2024; Cao et al., 2025a, Cao et al., 2025b). Under this role, the proposed hybrid agent-based and discrete-event simulation model can serve as an adaptive simulation platform that is dynamically calibrated using real-time data collected during an MCI. Such data may include emergency call reports, field triage updates, ambulance GPS information, road-condition updates, hospital information system data, and observed patient arrival patterns. As these data arrive over time, they can be used to update key simulation inputs and system parameters, such as casualty arrival rates, travel-time distributions, road availability, triage service times, and hospital-bed availability. Based on the recalibrated simulation model, the DNN-API algorithm can further update the post-decision-state value function and dynamically generate adaptive ambulance dispatch, hospital selection, and ambulance redeployment strategies as the disaster environment evolves. This role is particularly relevant to MCI environments because post-disaster conditions are highly unpredictable, nonstationary, and continuously changing over time. For example, unexpected casualty surges, sudden road disruptions, ambulance delays, or rapid changes in hospital capacity may cause the actual system to deviate from the scenarios used in offline training.

In this study, our implementation primarily focuses on the first role, namely the offline simulation–online application scheme. The hybrid simulation model is used to generate stochastic system transitions and reward samples offline, and the DNN-API algorithm uses these samples to train the value function approximation. The trained value function is then applied online to generate real-time operational decisions based on the currently observed MCI system state at each decision epoch. Therefore, the term "real-time decision-making" in this study refers to the fast online use of an offline-trained decision model, rather than continuous retraining during the disaster response process. Nevertheless, when sufficient streaming data become available in future applications, the proposed framework can be naturally extended to the second role, namely streaming-data-based adaptive real-time decision-making, in which the simulation model is dynamically recalibrated and the DNN-API policy is periodically updated in a rolling-horizon manner. Such an extension would allow the proposed framework to provide more adaptive and responsive decision support under evolving and nonstationary MCI conditions.

Beyond this streaming-data-based extension, future research can further explore the following important directions. First, it is essential to examine the impact of road repair decisions on stochastic dynamic ambulance dispatch strategies. In real post-earthquake scenarios, although some roads may be severely damaged, government agencies typically deploy repair crews promptly to restore critical road segments, aiming to reduce ambulance travel times and thereby improve casualty survival rates. Therefore, road repair and ambulance dispatch should be regarded as interdependent decision-making problems. Future work should focus on developing integrated decision models that can jointly optimize both aspects to enhance overall disaster response effectiveness. Second, as the determination of CCP locations represents a critical component of mass casualty incident management (Gormez et al., 2011), future research should investigate the simultaneous

optimization of CCP placement and dynamic ambulance dispatch and routing decisions. This line of inquiry aligns with the location-routing problem (LRP) in the logistics literature and has the potential to significantly enhance coordination and effectiveness in emergency response operations. Third, given the high computational complexity of solving the MDP model proposed in this study, future research should focus on developing alternative and more computationally efficient simulation-based ADP algorithms to enable scalable and practical solutions for large-scale stochastic dynamic ambulance dispatch and casualty distribution problems.

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CRediT authorship contribution statement

Kuo-Hao Chang: Writing – review & editing, Validation, Supervision, Project administration, Investigation, Conceptualization. **Tzu-Li Chen:** Writing – original draft, Validation, Supervision, Methodology, Investigation. **Lan-Hsin Yu:** Writing – review & editing, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation. **Tzu-Yin Chang:** Writing – review & editing, Validation, Supervision, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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